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# A hyper-heuristic based approach with naive Bayes classifier for the reliability *p*-median problem

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**Abstract**
Facility location models involve identifying locations for facilities that provide services to the customers which are also called demand points. The $p$-median problem is a facility location problem which deals with locating $p$ facilities in such a way that the sumtotal of demand-weighted distances between each demand point and its respective closest facility is minimized. The $p$-median problem does not take into account the possibility of failure of the facilities, but rather considers that the facilities once located will always be available to serve the customers or demand points. But, in reality, some of the facilities may face unpredicted disruptions, thereby forcing the customers to seek services from other functioning facilities. The reliability $p$-median problem (RpMP) concerns with locating facilities which minimize the cost while considering the cost of facility failures. In this paper, we have proposed a hyper-heuristic approach with naive Bayes classifier for the RpMP and compared its performance on standard benchmark instances with two best performing approaches for RpMP, viz. a genetic algorithm and a scatter search approach available in the literature. The results show the effectiveness of our approach in terms of the solution quality.

**Keywords** Hyper-heuristic · Naive Bayes classifier · $p$-median problem · Reliability $p$-median problem

## 1 Introduction

Facility location problems deal with identifying suitable locations for a set of facilities that serve a set of demand points or customers. Among the facility location problems, $p$-median problem (pMP) is a well studied problem which was introduced by [1] in 1964. The pMP is concerned with locating $p$ facilities in such a way that the sumtotal of demand-weighted distances between each demand point and its respective closest facility is minimized. It is an $\mathcal{NP}$-hard problem as shown by Kariv and Hakimi [2]. In the $p$-median problem, it is assumed that once constructed, the facilities will always operate as planned. But in reality, facilities may fail at times due to several reasons like natural disasters such as floods and earthquakes. Facilities may also fail due to events which are intentional like terrorist attacks and labor strikes. Sometimes facilities may fail due to unintended events like sudden power or component failures. When there are facility failures, there will be disruptions in the services provided to the customers. Depending on the type and severity of the events that make the facilities inoperable, the disruptions to the customer service can last for a short duration or for a longer period of time. In such facility failure cases, a customer who is generally serviced by the nearest facility now needs to be assigned to a distant functional facility. This increases the overall cost due to the additional distance the customer has to travel to a new facility to avail the services. For the first time, Snyder and Daskin [3] proposed the reliability $p$-median problem which introduced reliability approach in the facility location model taking facility failures into consideration. The reliability $p$-median problem (RpMP) minimizes both the primary transportation cost without considering the facility failures and also the cost of the expected failure considering the facility failures. The basic idea behind introducing the RpMP was to allot each customer to a primary facility which will serve it in normal situations, as well as to a set of backup facilities which will serve it in the case of failure of the primary facility. As multiple failures can happen at the same time, each customer

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requires a first backup facility when its primary facility fails, a second backup facility when its first backup facility fails, and so on. However, if a customer is allotted to a nonfailable facility as its $k^{th}$ backup, no more backups are required. Usually, nonfailable facilities are much less in numbers in comparison to failable ones owing to higher cost of the former. If all facilities are nonfailable then RpMP reduces to $p$-median problem. Owing to the provision of backup facilities at multiple levels, RpMP is a highly complex problem which can be used for modelling critical applications in different domains such as health care, aviation and defence [3].

Snyder and Daskin [3] while introducing RpMP also proposed an exact method for solving it based on Lagrangean relaxation. This method was suitable for solving small instances only. Later Alcaraz et al. [4] proposed eight hybrid metaheuristic approaches for RpMP based on genetic algorithm (GA) and scatter search (SS). Among these eight hybrid approaches, four (GA1, GA2, GA3, GA4) were based on GA and four (SS1, SS2, SS3, SS4) were based on SS. Among these eight approaches, GA2 and SS2 performed better than other approaches with SS2 performing slightly better than GA2. These are the only approaches available in the literature for RpMP. However, several problems which consider reliability issues in various kind of networks have been studied in the literature in the last few decades. A review of these problems have been provided in the next section.

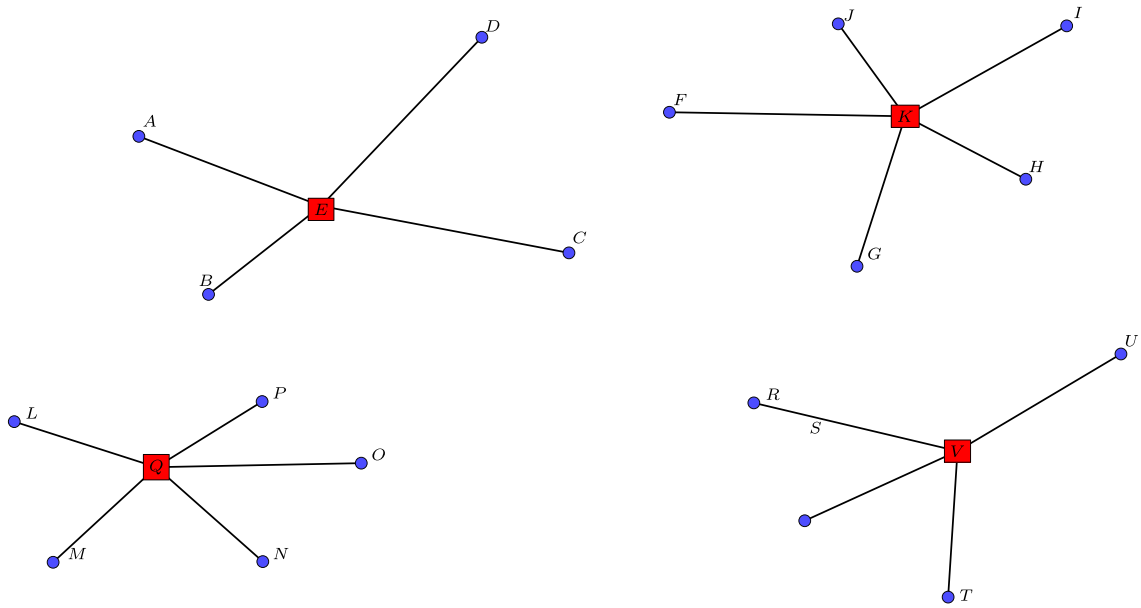
Consider an example network with 22 nodes and 4 facilities as depicted in Fig. 1. The nodes selected for facilities are shown in rectangle shape and marked in red color for easy identification. As shown in the Fig. 1a, nodes $\{A, B, C, D\}$ are assigned to their nearest facility $E$. Nodes $\{F, G, H, I, J\}$ are assigned to facility $K$. Similarly nodes $\{L, M, N, O, P\}$ are assigned to facility $Q$ and finally nodes $\{R, S, T, U\}$ are assigned to facility $V$. When the facility $E$ fails, out of the nodes assigned to it, $\{C, D\}$ are now reassigned to the next closest functioning facility which is $K$ as shown in Fig. 1b. And the nodes $\{A, B\}$ including the facility $E$ is now assigned to facility $Q$ which is the closest functioning facility. The re-assignment of nodes $\{A, B, C, D, E\}$ to the new facilities is depicted in dotted lines in Fig. 1b.

In this paper, we have proposed a hyper-heuristic based approach with naive Bayes classifier for solving the reliability $p$-median problem (RpMP). We have compared the results of our approach with the state-of-the-art methods available in the literature [4]. The effectiveness of the proposed approach can be observed from the superior quality of our solutions. Further, use of naive Bayes classifier not only reduces the total execution time, but also improves the solution quality. No approach exists in the literature so far that combines a machine learning technique with a hyper-heuristic approach for solving the facility location problems. Ours is the maiden approach combining the hyper-heuristic method and a machine learning technique for a facility location problem. This will serve as motivation for other researchers to develop analogous methods using other machine learning techniques including but not limited to naive Bayes classifier, support vector machine (SVM), reinforcement learning etc. for solving the different variety of facility location problems. Further, our approach can be easily adopted to other combinatorial optimization problems such as those presented in [5–10] by changing the set of low level heuristics as per the problem under consideration.

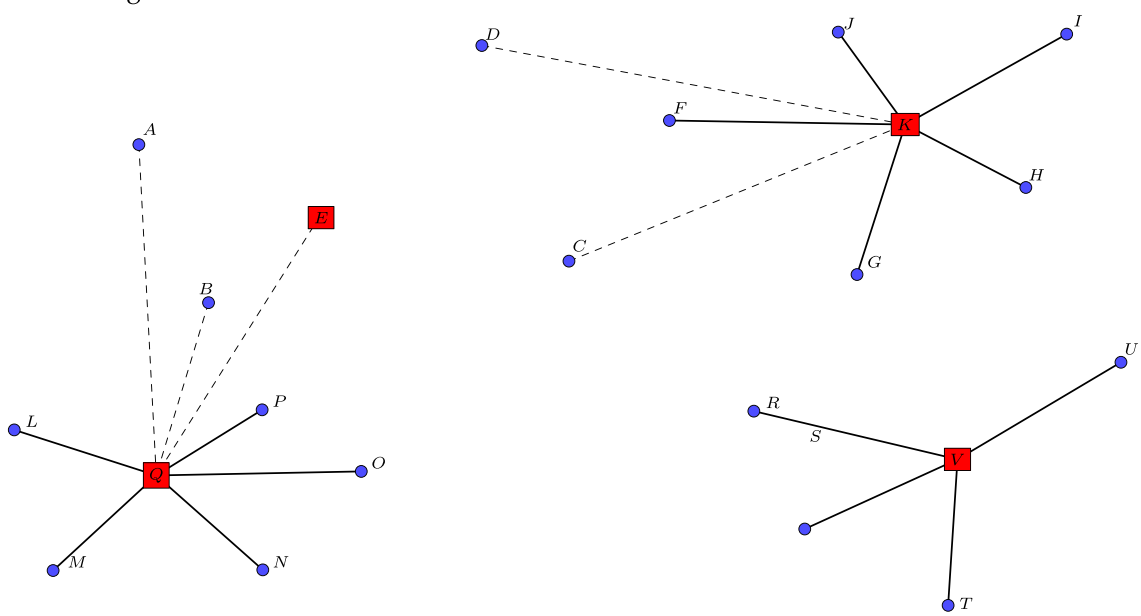
The remainder of this paper is organized as follows: Section 2 provides a literature review of the $p$-median problem and reliability $p$-median problem. Section 3 presents the formal problem definition of RpMP. Section 4 provides an introduction to hyper-heuristic approach, whereas Section 5 explains the functioning of naive Bayes with an example. Section 6 presents the proposed greedy selection based hyper-heuristic approach for RpMP. Experimental results and their analysis are presented in Section 7. Finally, Section 8 concludes the paper by listing the contributions made and a few directions for future research.

## 2 Literature review

Starting from the first formulation of the $p$-median problem provided in [11], there have been several different formulations of it over the years as mentioned in [12–14]. The capacitated $p$-median problem (CPMP) is a variant of pMP in which there is a capacity constraint on each of the possible facilities and each facility can only serve demand points within its capacity limit. The traditional pMP is an uncapacitated version where there is no capacity constraint on the facilities. Over the last few decades, several authors have proposed different methods to solve the pMP and its variants. Jayalakshmi and Singh [15] proposed a swarm intelligence approach using artificial bee colony algorithm to solve the pMP. In [13] Domínguez and Muñoz introduced a new reduced formulation for the pMP and proposed a recurrent neural network to solve this new formulation. In [14], Sourour proposed a tighter formulation of the pMP by mixed integer linear program and demonstrated that the standard branch-and-cut algorithm efficiently solves this tighter formulation on the considered benchmark instances . Keivan and Seyed [16] proposed a genetic algorithm based approach for the CPMP. Fleszar and Hindi [17] proposed a variable neighbourhood search based solution for the CPMP. In [18], Canós et al. introduced fuzzy $p$-median problem that allows some of the demand points to be uncovered if that gives significant lower cost. The fuzzy $p$-median problem finds its applications in private sector firms that try to maximize the profit while providing non-essential services and need not serve all the demand points. Canós et al. [18] also proposed an exact algorithm to solve this problem. In [19], Cadenas et al.




(a) Node assignments when all the four facilities are functioning




(b) Node assignments when one facility fails

**Fig. 1** Illustration of reliability $p$-median problem with $p = 4$ facilities

proposed a two-population genetic algorithm to solve the fuzzy $p$-median problem. The vector assignment $p$-median problem (VAPMP) [20] and distributed $p$-median problem [21] are the other variants of pMP in which the demand of a customer is collectively satisfied by different facilities at various levels and not just the closest facility.

Apart from the $p$-median problem and its variants, several authors studied reliability of telecommunication networks and power networks considering the failure of edges connecting the nodes [22–25]. These connectivity reliability theories deal with the probability of the network being connected even when there are failures along edges. There are several approaches proposed over the years to estimate the network reliability considering failures along the edges [26–29]. Not only the edge failures, taking the facility disruptions also into consideration, there have been several approaches proposed in the past to deal with network reliability issues [3, 30–35]. Unlike the vector assignment $p$-median problem (VAPMP),

in RpMP a customer is served by a level 2 facility only when the previous facility at level 1 fails. There have been location models introduced based on queueing of customers waiting to receive service from a facility in congested scenarios [36, 37]. Consolidating several methods of solving the facility location problems in systems with congestion, Berman and Krass [38] presented a complex model in an illustrative manner. RpMP differs from the facility location in congested systems as we deal with complete failure of a facility in RpMP as compared to a facility being unable to serve customers due to congestion. Afify et al. [39] and Afify et al. [40] studied the related facility location problems where there is only one layer of backup and limited budget is available to fortify a few facilities to make them non-failable. In comparison to these problems, the reliability model of RpMP introduced in [3] is way more complex and more relevant for critical applications.

## 3 Formal problem definition

To formally define RpMP, we have followed the notational conventions of [4]. Table 1 gives the summary of key notational conventions.

In the RpMP modeling, each customer $i \in A$ is either assigned to a failable facility or to a non-failable facility. A non-failable facility will never fail, and there is no need to consider re-assigning customer $i$ to any other facility. But if the customer $i$ is assigned to a failable facility then it is assigned to different facilities at different levels, viz. to an open facility at level 0, to a different facility at level 1 which will serve the customer $i$ if the facility at level 0 fails, and $i$ is assigned to yet another facility at level 2 when both the facilities at level 0 & level 1 fail, and so on till all the open facilities are considered. To represent the open facilities and the assignment of a customer to a facility at a level $r$, where $r = 0, \ldots, p-1$, we use two binary variables, viz. $S_j \in \{0, 1\}$, $W_{ijr} \in \{0, 1\}$. $S_j$, $W_{ijr}$ are referred to as location binary variable and assignment binary variable respectively. As the name suggests, $S_j = 1$ if there is an open facility at $j \in B$, otherwise $S_j = 0$. $W_{ijr} = 1$ if customer $i \in A$ is assigned to facility $j \in B$ at level $r$, otherwise $W_{ijr} = 0$.

Considering $\alpha \in [0, 1]$ and $R = \{0, 1, ..., p-1\}$, RpMP can be mathematically formulated in the following manner:

$$(RpMP) \;\; minimize \;\; \alpha c_1 + (1-\alpha)\, c_2 \tag{1}$$

subject to

$$\sum_{j \in B} W_{ijr} + \sum_{j \in NF} \sum_{x=0}^{r-1} W_{ijx} = 1 \;\; \forall i \in A, r \in R \tag{2}$$

$$W_{ijr} \leq S_j \;\; \forall i \in A, j \in B, r \in R \tag{3}$$

$$\sum_{j \in B} S_j = p \tag{4}$$

$$\sum_{r \in R} W_{ijr} \leq 1 \;\; \forall i \in A, j \in B \tag{5}$$

$$S_u = 1 \tag{6}$$

$$S_j \in \{0, 1\} \;\; \forall j \in B \tag{7}$$

$$W_{ijr} \in \{0, 1\} \;\; \forall i \in A, j \in B, r \in R \tag{8}$$

where

$$c_1 = \sum_{i \in A} \sum_{j \in B} h_i d_{ij} W_{ij0} \tag{9}$$

$$c_2 = \sum_{i \in A} h_i \Bigg[ \sum_{j \in NF} \sum_{r \in R} d_{ij} q^r W_{ijr} + \sum_{j \in F} \sum_{r \in R} d_{ij} q^r (1-q) W_{ijr} \Bigg] \tag{10}$$

Equation (1) gives the objective function of the problem which is the weighted sum of $c_1$ which is primary cost of serving customers assuming none of the facilities fail, and $c_2$ which is the expected failure cost considering the facility failures. Constraint in (2) makes sure that given a customer $i$ and level $r$, either $i$ is served by a facility $j \in B$ at level

**Table 1** Summary of key notations

| Notation | Meaning |
|---|---|
| $A$ | Set of demand points or customers |
| $B$ | Set of potential locations for facilities |
| $F$ | Subset of locations of candidate facilities($B$) which may fail |
| $NF$ | Subset of locations of candidate facilities($B$) which may not fail |
| $h_i$ | Demand at each customer $i \in A$ |
| $d_{ij}$ | Cost of service per unit of demand of customer $i \in A$ from facility $j \in B$ |
| $\theta_i$ | Cost per unit of demand of customer $i \in A$ when $i$ remains unserved |
| $q$ | Uniform probability of failure of each facility $j \in F$ |
| $p$ | Total number of facilities to be located |

$r$ or $i$ is served by a non-failable facility $j \in NF$ at level $x < r$. Constraint in (3) ensures that a customer $i$ can only be assigned to a location where a facility is located. Constraint in (4) requires that there are exactly $p$ facilities located. Constraint in (5) requires that a customer $i$ is assigned to a facility at one level at most. Constraint in (6) gives the information of an emergency facility being opened at an imaginary location $u$ which is non-failable, $u \in NF$. For each customer $i \in A$, if it remains unserved because it is not connected to any of the facilities or all the facilities have failed then it is considered that the customer $i$ is served by the emergency non-failable facility $u$ with a transportation cost of $d_{iu} = \theta_i$. (6), (7) give the binary nature of the location variables and assignment variables. Using (9), the primary cost of serving customers assuming none of the facilities fail $c_1$ is calculated, whereas using (10) the expected failure cost $c_2$ considering the facility failures is calculated.

## 4 Hyper-heuristic approach

To solve any $\mathcal{NP}$-hard combinatorial optimization problem, heuristic and metaheuristic approaches are usually developed by taking into account the problem specific information. To solve any new problem using the heuristic and metaheuristic approaches, one has to make major modifications in these approaches based on the characteristics of the problem to arrive at quality solutions in reasonable computational times. It is also shown in [41] that the quality of solutions generated by approaches combining various low-level heuristics in a proper way is better than those generated by each individual low-level heuristic. Hence, there arises the need for high-level solution methods without the problem specific knowledge that can be applied to various problems across domains to generate better quality solutions by combining the low-level heuristics. Denzinger et al. [42] first introduced the term *hyper-heuristic* as an automated theorem prover by combining various artificial intelligence techniques. But no formal definition was given for hyper-heuristics in [42]. Later Cowling et al. [43] described hyper-heuristics as the high-level heuristic to choose from a set of low-level heuristics.

Hyper-heuristics can be divided into two broad categories, namely generative hyper-heuristics and selective hyper-heuristics. The generative hyper-heuristics generate a new heuristic at each step of the search process and utilize the generated heuristic. The selective hyper-heuristics are those which select a heuristic from the available low-level heuristics at each step of the search process. In the recent years, hyper-heuristics have been used to solve numerous combinatorial optimization problems [44, 45]. But there is no hyper-heuristic based approach to solve the RpMP in the literature. This served as a motivation for us to propose a hyper-heuristic based approach for RpMP. In this paper, we have implemented a selective hyper-heuristic based approach which controls a set of low-level heuristics during an iterative search process to find high quality solutions.

## 5 Naive Bayes classifier

Naive Bayes classifier is a supervised classification algorithm. It is a probabilistic model and works according to the Bayes theorem. In the naive Bayes classifier, it is assumed that the features which are involved in classification are independent of each other given the class variable. Naive Bayes is a simple yet computationally efficient classifier which is unaffected by noise. Over the years, naive Bayes classifier has been extensively used for text classification [46, 47], in traffic risk management [48], to predict Alzheimer's disease from genome-wide data [49]. But no one so far applied naive Bayes to solve combinatorial optimization problems so as to improve the solution quality or reduce the execution time of the approaches. This served as a motivation for us to incorporate naive Bayes in our proposed hyper-heuristic approach.

Every supervised classifier has two major phases namely training phase and testing phase [50]. As part of the training phase, a prediction model is generated which correlates different features to a class label. In the naive Bayes classifier, at the end of the training phase, we calculate the class probability of each of the classes and we also calculate the conditional probabilities for each of the feature values given a class. We calculate the conditional probability of each feature value as ratio of the number of occurrences of that feature value for a given class to the total number of instances belonging to that class.

Probability of a feature vector $s$ belonging to a particular class $Y_j$ is calculated using Bayes theorem as follows:

$$p(Y_j|s) = \frac{p(Y_j)p(s|Y_j)}{p(s)} \tag{11}$$

Here $p(Y_j|s)$ is the posterior probability of $Y_j$ given the feature vector $s$. $p(Y_j)$ is the prior probability of class $Y_j$ which is calculated using the training data. $p(s|Y_j)$ is the conditional probability of the feature vector given the class $Y_j$. $p(s)$ is the prior probability of the feature vector.

The denominator in the (11), $p(s)$ has the same value for all the classes $j$, and hence, can be ignored in naive Bayes classifier while predicting the most likely class of a test sample. So, using the different probabilities which are calculated on the training data, the naive Bayes classifier predicts class label $Y$ for a test sample as below:

$$Y = \underset{j \in \{1,2,\ldots J\}}{\operatorname{argmax}} \; p(Y_j) \prod_{i=1}^{n} p(s_i|Y_j) \tag{12}$$

**Table 2** Sample dataset used for naive Bayes training

| S. No. | Feature values $s_1$ | $s_2$ | $s_3$ | Class label Y |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | $Y_1$ |
| 2 | 0 | 1 | 0 | $Y_2$ |
| 3 | 1 | 0 | 0 | $Y_1$ |
| 4 | 1 | 0 | 1 | $Y_2$ |
| 5 | 0 | 0 | 1 | $Y_1$ |

For example, consider a dataset of 5 training samples with three binary features $(s_1, s_2, s_3)$ and a class label $Y_j$ with two possible classes $Y_1$, $Y_2$ given in Table 2. In the 5 training samples, 3 samples belong to class $Y_1$ and 2 samples belong to class $Y_2$. At end of naive Bayes training phase, we calculate the class probabilities $p(Y_1)$ and $p(Y_2)$ as 0.6 and 0.4 respectively, which are proportional to the number of training samples belonging to the given class. In the 3 training samples belonging to class $Y_1$, let the feature value of $s_1$ is 1 in two samples and $s_1$ is 0 in one sample. Then we calculate the conditional probabilities $p((s_1 = 1)|Y_1)$ and $p((s_1 = 0)|Y_1)$ as $\frac{2}{3}$ and $\frac{1}{3}$ respectively. Similarly, in the two samples belonging to class $Y_2$, the feature value of $s_1$ is 1 in one sample and $s_1$ is 0 in the other sample. Then we calculate conditional probabilities $p((s_1 = 1)|Y_2)$ and $p((s_1 = 0)|Y_2)$ as $\frac{1}{2}$ and $\frac{1}{2}$ respectively. The conditional probabilities for all the three features $s_1$, $s_2$ and $s_3$ for their corresponding feature values are given in Table 3. Given a test sample $s$, with naive Bayes we need to predict whether it belongs to the class $Y_1$ or class $Y_2$ using the conditional probabilities of its features $(s_1, s_2, s_3)$ and the class probabilities $p(Y_1)$, $p(Y_2)$. Let us consider a test sample $s = (0, 1, 1)$. As per naive Bayes classifier, we calculate the posterior probabilities of the given test sample belonging to class $Y_1$, $Y_2$ as $p(Y_1|s = (0, 1, 1))$, $p(Y_2|s = (0, 1, 1))$, where $p(Y_1|s = (0, 1, 1)) = p(Y_1) * p(s_1 = 0|Y_1) * p(s_2 = 1|Y_1) * p(s_3 = 1|Y_1)$ and $p(Y_2|s = (0, 1, 1)) = p(Y_2) * p(s_1 = 0|Y_2) * p(s_2 = 1|Y_2) * p(s_3 = 1|Y_2)$. From the previously computed conditional probabilities and class probabilities using the training data, we can compute $p(Y_1|s = (0, 1, 1)) = 0.6 * \frac{1}{3} * \frac{1}{3} * \frac{1}{3} = 0.022$. Similarly we can compute $p(Y_2|s = (0, 1, 1)) = 0.4 * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} = 0.05$.

**Table 3** Conditional probabilities for different feature values

| Feature | Class $Y_1$ $p(s_i = 0\|Y_1)$ | $p(s_i = 1\|Y_1)$ | Class $Y_2$ $p(s_i = 0\|Y_2)$ | $p(s_i = 1\|Y_2)$ |
|---|---|---|---|---|
| $s_1$ | $\frac{1}{3}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |
| $s_2$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |
| $s_3$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |

Since $p(Y_2|s = (0, 1, 1)) > p(Y_1|s = (0, 1, 1))$, naive Bayes predicts that the given sample belongs to class $Y_2$.

## 6 Proposed approach

This section presents the salient features of our hyper-heuristic approach.

### 6.1 Solution representation and fitness

We represent each solution as a set of $p$ nodes to locate facilities. The objective function given in (1) is used as the fitness function, which is the weighted sum of the cost of serving customers when there are no facility failures and the expected failure cost which includes the cost of serving customers when there are facility failures.

### 6.2 Generating the initial solution

We have used two methods to generate the initial solutions, viz. a randomized greedy approach in the first iteration of the hyper-heuristic and a random generation method for generating the initial solutions in all the remaining iterations of the hyper-heuristic. In the randomized greedy approach to generate the initial solution, we have made use of the node connectivity of the given input network. In a network, which is an incomplete graph where a given node is not connected to every other node, the facility located at a node can serve only the set of nodes it is connected to. Making use of this nature of the input network, as part of the randomized greedy approach, first we select five nodes with the highest number of adjacent nodes and randomly select one node from these five nodes, and make it part of the solution. The selected node is removed from the list of nodes available for locating facilities. This previous step is repeated till we select a total of $p$ nodes to be part of the initial solution. For the remaining restart iterations of the hyper-heuristic, we generate the initial solutions in a random manner by randomly selecting $p$ unique nodes to be part of the initial solution. In both the randomized greedy approach and the random generation method of generating the initial solution, there may be some customers which are not connected to any of the $p$ facilities, and hence, remain unserved. For all such unserved customers $i$, like [4], we have used a fixed cost $\theta_i$ as the cost of serving while calculating the fitness value.

### 6.3 Hyper-Heuristic framework with naive Bayes classifier

We have implemented a greedy selection based hyper-heuristic approach with multiple starts. In every start, we generate an initial solution, and then apply four low level

heuristics in two phases namely training phase and testing phase. The training phase and testing phase are part of the naive Bayes classifier which is deployed in this case to reduce the computation time while improving the solution quality. The four low level heuristics that we have used are described in the subsequent subsection (Section 6.4). We will refer to our four low level heuristics as $LH_1$, $LH_2$, $LH_3$, and $LH_4$ subsequently.

In an iteration of the **training phase**, all the four low level heuristics are applied on the current solution $X_{tr}$ one-by-one thereby creating four new solutions. We have used variable degree of perturbation while applying each of the low level heuristics over many training iterations which varies from $Max_{prt}$, which is the maximum degree of perturbation, to $Min_{prt}$, which is the minimum degree of perturbation over the maximum number of training iterations, $Max_{tr}$. To generate good solutions, we need higher perturbation in the initial iterations and lower perturbation towards the final iterations. In training iteration $iter_{tr}$, the degree of perturbation is calculated as $Deg_{prt} = \left(\frac{Max_{prt}-Min_{prt}}{Max_{tr}}\right)(Max_{tr} - iter_{tr}) + Min_{prt}$ [51]. We select the solution with the least objective value among the four resulting solutions after applying the four heuristics as an input to the next iteration of the training phase. As part of the naive Bayes classification, we treat each of the four low level heuristics as four different classes and in one training phase iteration which ever low level heuristic produces least objective solution, it is considered that the given solution $X_{tr}$ belongs to that particular class. At the end of the training phase, we find the class probabilities of each of the four low level heuristics based on the number of times a low level heuristic generates the solution with least objective value. We calculate the conditional probabilities of feature values of the solution used for for training, $X_{tr}$, considering a binary value of 1 for all the nodes which are part of the solution and a binary value of 0 for all the nodes which are not part of the solution. In the next paragraph, we have explained the training phase with the help of an example.

Consider a network with 7 nodes and the number of facilities to be located $p = 3$. We maintain a two dimensional array $Node_cnt[NH][N]$, where $NH$ and $N$ are the total number of low level heuristics and the total number of nodes in the network respectively. At the end of the training phase, the value at $Node_cnt[i][j]$ represents the number of times node $j$ is part of the solution when the low level heuristic $LH_i$ has generated a better solution among all the low level heurstics. All the cells in the matrix $Node_cnt[NH][N]$ are assigned with zeros in the beginning. Let the initial solution generated to this problem be $X_{init} = \{2, 4, 6\}$ which means out of the 7 nodes in the network $2^{nd}$, $4^{th}$ and $6^{th}$ node are selected for locating facilities. Consider that the training phase is applied for 6 iterations. On applying all the four low level heuristics on $X_{init}$, let us say the first low level heuristic $LH_1$ generated a better solution among the four low level heuristics. So, we increment the values in $Node_cnt[1][2]$, $Node_cnt[1][4]$, $Node_cnt[1][6]$ by one. Let $X_{tr} = \{1, 2, 3\}$ be the resulting solution after applying $LH_1$ on $X_{init}$. Now $X_{tr}$ works as an input to the second iteration of the training phase. Consider that $LH_2$ generated a better solution in the second iteration of training phase. Then we increment the values in $Node_cnt[2][1]$, $Node_cnt[2][2]$, $Node_cnt[2][3]$ by one. Let $X_{tr} = \{4, 5, 6\}$ be the resulting solution at the end of $2^{nd}$ iteration, which will now be given as input to the $3^{rd}$ iteration of training phase. Consider that $LH_2$ generated a better solution in the third iteration of training phase and we increment the values in $Node_cnt[2][4]$, $Node_cnt[2][5]$, $Node_cnt[2][6]$ by one. Let $X_{tr} = \{2, 3, 7\}$ be the resulting solution at the end of $3^{rd}$ iteration and it is the input to $4^{th}$ iteration of training phase. Considering $LH_3$ performs better in the $4^{th}$ iteration, the values in $Node_cnt[3][2]$, $Node_cnt[3][3]$, $Node_cnt[3][7]$ are incremented by one. The resulting solution from $4^{th}$ iteration which is an input to $5^{th}$ iteration be $\{1, 4, 6\}$. Assume $LH_3$ and $LH_4$ performed better in the $5^{th}$ and $6^{th}$ iterations generating the solutions $\{1, 5, 7\}$ and $\{1, 2, 4\}$ respectively. So, we increment the corresponding values in $Node_cnt[3][1]$, $Node_cnt[3][4]$, $Node_cnt[3][6]$ and $Node_cnt[4][1]$, $Node_cnt[4][5]$, $Node_cnt[4][7]$ by one. We presented the information of the six training phase iterations in Table 4. The first column in Table 4 titled $Iteration$ gives the iteration number. The second column, **Input sol.**, gives the input solution that is considered in that particular iteration. The third column, $Hr$, gives the low level heuristic that generated best resulting solution after all the four low level heuristics are applied on the input solution. The fourth column titled $Node_cnt[Hr][]$ gives the updated row values of the $Node_cnt$ matrix corresponding to the best performing heuristic. The fifth and final column titled **Resulting sol.** gives the resulting solution obtained by applying the best performing heuristic given in third column on the input solution given in second column. As previously explained, the resulting solution of first iteration is given as input to the

**Table 4** Training data information

| Iteration | Input sol. | Hr | $Node_cnt[Hr][]$ | Resulting sol. |
|---|---|---|---|---|
| 1 | $\{2, 4, 6\}$ | $LH_1$ | $\{0, 1, 0, 1, 0, 1, 0\}$ | $\{1, 2, 3\}$ |
| 2 | $\{1, 2, 3\}$ | $LH_2$ | $\{1, 1, 1, 0, 0, 0, 0\}$ | $\{4, 5, 6\}$ |
| 3 | $\{4, 5, 6\}$ | $LH_2$ | $\{1, 1, 1, 1, 1, 1, 0\}$ | $\{2, 3, 7\}$ |
| 4 | $\{2, 3, 7\}$ | $LH_3$ | $\{0, 1, 1, 0, 0, 0, 1\}$ | $\{1, 4, 6\}$ |
| 5 | $\{1, 4, 6\}$ | $LH_3$ | $\{1, 1, 1, 1, 0, 1, 1\}$ | $\{1, 5, 7\}$ |
| 6 | $\{1, 5, 7\}$ | $LH_4$ | $\{1, 0, 0, 0, 1, 0, 1\}$ | $\{1, 2, 4\}$ |

second iteration and the resulting solution from second iteration is given as input to the third iteration and so on, which can be observed from the Table 4. The final result after the last iteration of the training phase acts as input to the test phase.

$$Node_cnt = \begin{bmatrix} 0\,1\,0\,1\,0\,1\,0 \\ 1\,1\,1\,1\,1\,1\,0 \\ 1\,1\,1\,1\,0\,1\,1 \\ 1\,0\,0\,0\,1\,0\,1 \end{bmatrix}$$

Before starting the test phase, we calculate the class probabilities for the four low level heuristics and also the conditional probabilities for the feature values. Out of the 6 iterations for the training phase, as explained previously each of $LH_1$ and $LH_4$ generated better solution once, and, each of $LH_2$ and $LH_3$ generated better solutions twice. The class probabilities of $LH_1$ and $LH_4$ are $p(LH_1) = \frac{1}{6}$, $p(LH_4) = \frac{1}{6}$. Similarly the class probabilities of $LH_2$ and $LH_3$ are $p(LH_2) = \frac{2}{6}$, $p(LH_3) = \frac{2}{6}$. Using the $Node_cnt$, matrix we can calculate the conditional probabilities for each node being part of the solution and not being part of the solution as given in the Table 5. Now, the resulting solution at the end of the $6^{th}$ iteration of the training phase, $X_{tr} = \{1, 2, 4\}$ is given as input to the test phase $X_{test}$ and then the testing phase starts.

In each iteration of the **testing phase**, for the given solution $X_{test}$, we find the posterior probabilities for each of the four low level heuristics using naive Bayes classifier, $p(LH_1|X_{test}), p(LH_2|X_{test}), p(LH_3|X_{test}), p(LH_4|X_{test})$. We will skip the low level heuristic with the least posterior probability. For the above example, the input to the test phase is $X_{test} = \{1, 2, 4\}$, and we calculate the posterior probability of $LH_1$ as $p(LH_1|X_{test}) = p(LH_1) * p(n_1 = 1|LH_1) * p(n_2 = 1|LH_1) * p(n_3 = 0|LH_1) * p(n_4 = 1|LH_1) * p(n_5 = 0|LH_1) * p(n_6 = 0|LH_1) * p(n_7 = 0|LH_1) = \frac{1}{6} * 0 * 1 * 1 * 1 * 1 * 0 * 1 = 0$. Similarly, we calculate the $p(LH_2|X_{test})$, $p(LH_3|X_{test})$ and $p(LH_4|X_{test})$ as 0.0052, 0.0052 and 0 respectively. Since the posterior probabilities of both $LH_1$ and $LH_4$ are 0, $LH_1$ which comes first in the indexing order is skipped and the remaining three low level heuristics $LH_2$, $LH_3$ and $LH_4$ are applied on the solution $X_{test}$. The degree of perturbation in a test iteration $iter_{test}$ is calculated as $Deg_{prt} = \left(\frac{Max_{prt} - Min_{prt}}{Max_{test}}\right)(Max_{test} - iter_{test}) + Min_{prt}$, where $Max_{test}$ is the maximum number of testing phase iterations. Then we select the solution with the least objective value among the three resulting solutions after applying the three heuristics. Then on the selected solution, we apply the 1-1 exchange local search operation which is explained the subsequent subsection, viz. Section 6.5. The resulting solution after local search is considered as an input to the next iteration of the testing phase. If there is no improvement in the best solution objective value for five consecutive test iterations, we stop the testing phase. Once the testing phase ends, we apply 1-1 exchange on the best solution found so far and then the hyper-heuristic makes a fresh start. The pseudo-code of our approach is presented in Algorithm 1.

There are several acceptance criterias discussed in the hyper-heuristic literature [52] for selecting the newly generated solution as current solution in the next iteration. In our proposed approach, we have experimented with two acceptance criterias namely AA (all acceptance), OI (only improvement). The AA criteria always selects the newly generated solution, whereas the OI criteria selects the newly generated solution only if it is better than the current solution. In our experiments, we have found that the AA criteria provided better results in comparison to the OI criteria. This can be attributed to the better exploration of the search space as a result of following the AA criteria due to the selection of a new solution in every iteration. Hence, we have used the AA criteria in both the training and testing phases of our proposed hyper-heuristic approach.

### 6.4 Low level heuristics

We have used the following four low level heuristics in our hyper-heuristic approach:

**Table 5** Conditional probabilities for different feature values

| Node | Class $LH_1$ | | Class $LH_2$ | | Class $LH_3$ | | Class $LH_4$ | |
|---|---|---|---|---|---|---|---|---|
| | $p(n_i = 0\|LH_1)$ | $p(n_i = 1\|LH_1)$ | $p(n_i = 0\|LH_2)$ | $p(n_i = 1\|LH_2)$ | $p(n_i = 0\|LH_3)$ | $p(n_i = 1\|LH_3)$ | $p(n_i = 0\|LH_4)$ | $p(n_i = 1\|LH_4)$ |
| $n_1$ | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | 1 |
| $n_2$ | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 0 |
| $n_3$ | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 0 |
| $n_4$ | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 0 |
| $n_5$ | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 0 | 0 | 1 |
| $n_6$ | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 0 |
| $n_7$ | 1 | 0 | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | 1 |

**Algorithm 1** Greedy hyper-heuristic approach with naive Bayes classifier for the RpMP.

```
Input: List of parameters for greedy hyper-heuristic and an
instance of RpMP
Output: Overall best solution found
X_best ← ∅ ;
while termination condition remains unsatisfied do
    X_init ← Generate initial solution;
    X_tr ← X_init ;
    while (iter_tr ≤ Max_tr) do
        Deg_prt ←
        ((Max_prt − Min_prt) / Max_tr)(Max_tr − iter_tr) + Min_prt;
        X_1 ← LH_1(X_tr, Deg_prt);
        X_2 ← LH_2(X_tr, Deg_prt);
        X_3 ← LH_3(X_tr, Deg_prt);
        X_4 ← LH_4(X_tr, Deg_prt);
        X_tr ← MIN(X_1, X_2, X_3, X_4);
        if (X_tr is better than X_best) then
            X_best ← X_tr;
    X_test ← X_tr ;
    while (iter_test ≤ Max_test) do
        skip_hr ← Naive_Bayes(X_test);
        Deg_prt ←
        ((Max_prt − Min_prt) / Max_test)(Max_test − iter_test) + Min_prt;
        for (i := 1 to 4) do
            if (skip_hr! = i) then
                X_i ← LH_i(X_test, Deg_prt);
        Fitness(X_skip_hr) ← RAND_MAX;
        X_test ← MIN(X_1, X_2, X_3, X_4);
        X_test ← Local_search(X_test);
        if (X_test is better than X_best) then
            X_best ← X_test;
    X_best ← Local_search(X_best);
return X_best;
```

- $LH_1$**: Random removal and greedy addition**
  In the first low level heuristic, from the given solution $sol$ containing $p$ facilities, a given number $NR(= p * Deg_{prt})$ facilities are removed randomly. Now, to again get a feasible solution with $p$ facilities, we need to add $NR$ nodes back to the solution. These $NR$ nodes are added in an iterative manner where during each iteration we select a node with the maximum adjacency count from all the nodes which are not currently part of the solution, and add it to the solution.
- $LH_2$**: Random addition and greedy removal**
  As part of the second low level heuristic, from the list of nodes which are not part of the given solution $sol$ containing $p$ facilities, a fixed number $NA(= p*Deg_{prt})$ of nodes are randomly selected and added to the current solution. Now, the solution contains $p + NA$ facilities. To remove the extra $NA$ facilities from $sol$, we eliminate one facility at a time from the solution $sol$ in an iterative manner till there are only $p$ facilities remaining. In each iteration, the facility with minimum adjacency count is chosen for removal.
- $LH_3$**: Greedy removal and greedy addition**
  In the third low level heuristic, from the given solution with $p$ facilities, we iteratively select a facility with the least adjacency count and remove it from the solution. In this fashion, a total of $NR(= p * Deg_{prt})$ facilities are removed from the given solution. Then, from all the locations which are not part of the solution, we select a node with the maximum adjacency count and add it to the solution. In this manner, nodes are added iteratively till the total number of facilities in the solution reaches $p$.
- $LH_4$**: Greedy addition and greedy removal**
  In the fourth low level heuristic, to the given solution having $p$ facilities, we add $NA(= p*Deg_{prt})$ new facilities in a greedy manner. This is an iterative procedure where in each iteration, out of all the locations which are not part of the solution, we select a location with the maximum adjacency count and add it to the solution. At the end of this iterative process, the solution contains $p+NA$ facilities. Then, we select a facility with the least adjacency count and remove it from the solution. This process is repeated and is stopped when there are only $p$ facilities remaining in the solution.

### 6.5 Local search

As part of the local search, we have implemented a 1-1 exchange where for each facility in the given solution, we try to find a replacement node from the set of nodes which are not part of the solution. For this, a facility is considered for replacement with all $N - p$ nodes which are not part of the solution and is replaced with the node that provides the solution with the least objective value [53]. A facility which is replaced from the solution with a new node, can be reintroduced into the solution at a different index. For a facility under consideration if there is no replacement node available whose addition to the solution in place of the given facility results in a reduction in the objective value, then that facility continues to be part of the solution and exchange of that facility does not happen. And the next facility in the solution is considered for replacement. This iterative process is repeated for all $p$ facilities in the solution. Within the testing phase, we apply the 1-1 exchange for the initial two test iterations and also when the current solution objective value is less than the current best solution objective value. But for the cases when the current solution objective value is greater than that of the best solution, we apply the 1-1 exchange only if the difference is within 20% of the best solution's objective value. We have arrived at this 20% after a large number of experiments.

**Table 6** Comparison of results of hyper-heuristic with GA2 and SS2 (DLP Library)

| | | %Deviation | | | | CPU Time | | | |
|---|---|---|---|---|---|---|---|---|---|
| N | p | GA2 | SS2 | HH(without NB) | HH(with NB) | GA2 | SS2 | HH(without NB) | HH(with NB) |
| 50 | 5 | 0.77 | 1.01 | **0** | **0** | 0.33 | 0.29 | 0.39 | 0.41 |
| 50 | 10 | 2.06 | 0.48 | **0** | 0.21 | 8.21 | 0.56 | 3.03 | 1.96 |
| 50 | 15 | 0.36 | 0.36 | **0.06** | **0.06** | 2.27 | 1.1 | 5.56 | 3.76 |
| TOTAL 50 | | 1.07 | 0.62 | **0.02** | 0.09 | 3.61 | 0.65 | 2.99 | 2.04 |
| 100 | 5 | 0.37 | 0.46 | **-1.02** | **-1.02** | 1.82 | 1.54 | 1.67 | 1.05 |
| 100 | 10 | -0.43 | -1.22 | **-3.59** | -3.55 | 12.46 | 6.6 | 16.00 | 12.82 |
| 100 | 15 | -1.23 | -0.95 | -2.48 | **-2.93** | 20.47 | 18.4 | 29.42 | 33.63 |
| TOTAL 100 | | -0.43 | -0.57 | -2.36 | **-2.5** | 11.59 | 8.85 | 15.70 | 15.83 |
| OVERALL | | 0.32 | 0.02 | -1.17 | **-1.21** | 7.60 | 4.75 | 9.35 | 8.94 |

## 7 Computational results

We have implemented the proposed hyper-heuristic approach in C. In all our experiments, we have taken the maximum number of training iterations, $Max_{tr} = 15$ and the maximum number of test iterations, $Max_{test} = 10$. We have taken the maximum perturbation $Max_{prt} = 0.4$, and minimum perturbation $Min_{prt} = 0.1$. We have chosen these parameter values empirically after a large number of experiments. The cost of not serving a customer $i$ by any facility, $\theta_i$ is taken as 300.0 just as used in [4]. We have also conducted our experiments with two values of $\alpha$, 0.2 and 0.8. We have considered the same failure probability $q = 0.05$ and the demand of each customer $h_i \forall i \in A$ is taken as 1 just like in [4].

We have compared the results of our approach with the state-of-the-art methods for RpMP proposed in [4]. Just like in [4], we have also executed our approach on the same two datasets namely Discrete Location Problems Library (DLPL)[1] and the OR library[2] which Beasley [54] introduced. We have considered the termination condition of our approach as the maximum execution time in seconds which is a combination of the number of nodes $n$ and the number of facilities to be located $p$ and is calculated as $2 * \ln \binom{n}{p}$, just like in [4]. We have executed our hyper-heuristic approach on a Linux based 3.10 GHz Core-i5-8600 system with 8 GB RAM. We have reported the results of our approach with the naive Bayes classifier and without the naive Bayes classifier also to show the benefit of former over latter.

As part of the experimentation in [4], they have solved all the instances using CPLEX executing each instance for a maximum cpu time of 2 hours. In allotted time of two hours, for some instances CPLEX found the optimal solution, for some CPLEX found a feasible solution and for some CPLEX even fails to find even any feasible solution. To evaluate the performance of genetic algorithm and scatter search approaches proposed in [4], they have calculated the percentage deviation of the objective values of their approaches from the solution obtained by CPLEX at the end of two hours if CPLEX was able to find any feasible solution. Otherwise, the deviation is computed from the best solution among all the 8 genetic algorithm and scatter search based approaches. Out of the total 8 variations of the methods proposed in [4], they have reported GA2 and SS2 to be the best performing methods. We have obtained the instance-by-instance results from the first author of [4] through personal communication, and reported on each instance, the percentage deviation of the objective value of our hyper-heuristic approach with respect to the objective value of the solution used as reference in [4] for that instances. We have compared our results with the results of only GA2 and SS2 which are shown to be the best performing methods in [4]. Table 6 gives the comparison of results on DLPL instances, Table 7 gives the comparison of results on OR library instances where CPLEX has reached a feasible solution, and Table 8 gives the comparison of results on OR library instances where CPLEX has not reached a feasible solution. In all of the three tables, viz. Tables 6, 7, and 8, first column $N$ represents the number of nodes in the network, second column $p$ gives the number of facilities to be located. The values in each column under the heading %Deviation report the the percentage deviation of the corresponding approach's objective value from the objective value of the optimal/best solutions obtained by CPLEX on the same datasets. The 3rd, 4th, 5th and 6th columns give the percentage deviation of the objective values obtained by GA2, SS2, hyper-heuristic without naive Bayes and hyper-heuristic with naive Bayes respectively. The values in columns under the heading CPU Time, i.e., values in the 7th, 8th, 9th and 10th columns provide the time taken in seconds to arrive at the

[1] http://math.nsc.ru/AP/benchmarks/UFLP/

[2] http://people.brunel.ac.uk/~mastjjb/jeb/orlib/esteininfo.html

**Table 7** Comparison of results of hyper-heuristic with GA2 and SS2 (OR Library: Instances where CPLEX has reached a feasible solution)

| | | %Deviation | | | | CPU Time | | | |
|---|---|---|---|---|---|---|---|---|---|
| N | p | GA2 | SS2 | HH(without NB) | HH(with NB) | GA2 | SS2 | HH(without NB) | HH(with NB) |
| 100 | 5 | **0** | **0** | **0** | **0** | 0.08 | 0.13 | 0.02 | 0.01 |
| 100 | 10 | **0** | 0.03 | **0** | **0** | 2.3 | 0.52 | 1.49 | 0.92 |
| 100 | 20 | **0** | 0.43 | 1.25 | **0** | 45.42 | 13.62 | 44.60 | 23.15 |
| 100 | 20 | 0.99 | 0.91 | 1.31 | **0.81** | 45.62 | 4.36 | 30.59 | 30.85 |
| 100 | 33 | 0.14 | 0.46 | **0.02** | **0.02** | 33.09 | 10.18 | 77.64 | 72.05 |
| TOTAL 100 | | 0.22 | 0.37 | 0.52 | **0.17** | 25.30 | 5.76 | 30.87 | 25.4 |
| 200 | 5 | **0** | **0** | **0** | **0** | 0.42 | 0.50 | 0.05 | 0.03 |
| 200 | 10 | **0** | 0.04 | **0** | **0** | 18.93 | 6.59 | 5.42 | 13.41 |
| 200 | 20 | -9.74 | -9.88 | **-9.97** | **-9.97** | 103.36 | 47.32 | 24.84 | 14.57 |
| 200 | 40 | 6.22 | 1.78 | 1.56 | **1.33** | 174.7 | 92.76 | 81.47 | 95.00 |
| 200 | 67 | 0.9 | **0.29** | 0.38 | 0.91 | 206.21 | 90.94 | 110.68 | 113.86 |
| TOTAL 200 | | -0.52 | -1.55 | **-1.61** | -1.55 | 100.72 | 47.62 | 44.49 | 47.37 |
| 300 | 5 | **0** | **0** | **0** | **0** | 0.76 | 1.11 | 112.09 | 0.06 |
| 300 | 10 | **0** | **0** | **0** | **0** | 28.69 | 4.98 | 52.52 | 3.33 |
| 300 | 30 | -83.74 | **-84.75** | -84.55 | -84.54 | 146.6 | 135.67 | 112.09 | 66.11 |
| 300 | 60 | -21.4 | -31.67 | -31.17 | **-32.23** | 238.9 | 227.32 | 265.88 | 257.56 |
| 300 | 100 | 14.08 | 0.38 | 1.92 | **0.21** | 334.84 | 339.61 | 341.88 | 370.53 |
| TOTAL 300 | | -18.21 | -23.21 | -22.76 | **-23.31** | 149.96 | 141.73 | 176.89 | 139.52 |
| 400 | 5 | **0** | **0** | **0** | **0** | 3.45 | 2.05 | 0.19 | 0.11 |
| 400 | 10 | **-0.27** | -0.25 | -0.26 | -0.26 | 37.1 | 15.10 | 11.33 | 39.78 |
| 400 | 133 | -9.98 | -11.89 | -14.43 | **-14.92** | 349.4 | 559.20 | 725.89 | 578.74 |
| TOTAL 400 | | -3.42 | -4.05 | -4.90 | **-5.06** | 129.98 | 192.12 | 245.80 | 206.21 |
| 500 | 5 | **0** | **0** | **0** | **0** | 3.66 | 3.2 | 0.29 | 0.17 |
| OVERALL | | -5.41 | -7.06 | -7.05 | **-7.3** | 93.34 | 81.85 | 105.21 | 88.43 |

**Table 8** Comparison of results of hyper-heuristic with GA2 and SS2 (OR Library: Instances where CPLEX has not reached a feasible solution)

| | | %Deviation | | | | CPU Time | | | |
|---|---|---|---|---|---|---|---|---|---|
| N | p | GA2 | SS2 | HH(without NB) | HH(with NB) | GA2 | SS2 | HH(without NB) | HH(with NB) |
| 400 | 40 | 23.57 | 2.59 | 5.80 | **0.95** | 202.77 | 211.21 | 189.09 | 146.20 |
| 400 | 80 | 28.83 | 1.18 | **0.48** | 1.08 | 281.02 | 375.76 | 426.81 | 261.49 |
| TOTAL 400 | | 26.20 | 1.88 | 3.14 | **1.01** | 241.9 | 293.48 | 307.95 | 203.85 |
| 500 | 10 | 0.13 | **0** | 0.24 | 0.24 | 69.35 | 43.37 | 75.24 | 12.38 |
| 500 | 50 | 10.07 | 1.52 | **-0.09** | 0.04 | 260.82 | 311.99 | 269.57 | 322.40 |
| 500 | 100 | 27.25 | 0.45 | **-3.33** | -1.87 | 286.73 | 533.27 | 651.79 | 621.57 |
| 500 | 167 | 6.99 | **1.82** | 4.51 | 4.04 | 561.21 | 654.99 | 1037.61 | 859.36 |
| TOTAL 500 | | 11.11 | 0.95 | **0.33** | 0.61 | 294.53 | 385.9 | 508.55 | 453.92 |
| 600 | 5 | **0** | **0** | **0** | **0** | 3.48 | 4.44 | 1.22 | 0.48 |
| 600 | 10 | 0.46 | 0 | **-0.01** | **-0.01** | 58.20 | 19.87 | 24.03 | 1.52 |
| 600 | 60 | 14.37 | 3.13 | -0.88 | **-2.45** | 264.49 | 407.79 | 448.91 | 311.17 |
| 600 | 120 | 25.53 | 1.94 | 2.49 | **-1.71** | 525.04 | 677.22 | 754.92 | 1174.38 |
| 600 | 200 | 24.91 | 1.02 | **-0.24** | -0.14 | 1021.13 | 977.4 | 2451.07 | 2078.27 |

**Table 8** continued

| N | p | %Deviation | | | | CPU Time | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | GA2 | SS2 | HH(without NB) | HH(with NB) | GA2 | SS2 | HH(without NB) | HH(with NB) |
| TOTAL 600 | | 13.05 | 1.22 | 0.27 | **-0.86** | 374.47 | 417.34 | 736.03 | 713.16 |
| 700 | 5 | 0 | **0** | 0.01 | 0.01 | 5.48 | 6.09 | 0.58 | 0.33 |
| 700 | 10 | 0.2 | **0** | 0.26 | 0.71 | 72.63 | 23.66 | 13.83 | 87.43 |
| 700 | 70 | 11.38 | 4.76 | 0.95 | **-0.02** | 366.20 | 461.3 | 603.92 | 535.99 |
| 700 | 140 | 30.47 | 0.82 | **0.78** | **0.78** | 744.83 | 921.46 | 1526.23 | 1227.65 |
| TOTAL 700 | | 10.51 | 1.39 | 0.50 | **0.37** | 297.28 | 353.13 | 536.14 | 462.85 |
| 800 | 5 | **0** | **0** | **0** | **0** | 9.13 | 7.86 | 4.31 | 1.28 |
| 800 | 10 | 0.34 | 0.01 | **0** | **0** | 72.55 | 38.89 | 9.09 | 5.34 |
| 800 | 80 | 23.69 | 0.9 | 3.62 | **-1.99** | 370.13 | 590.44 | 538.19 | 787.29 |
| TOTAL 800 | | 8.01 | 0.3 | 1.21 | **-0.66** | 150.6 | 212.39 | 183.86 | 264.64 |
| 900 | 5 | 0 | 0 | **-0.01** | **-0.01** | 14.54 | 11.13 | 0.94 | 0.55 |
| 900 | 10 | 0.45 | **0.13** | 0.24 | 0.22 | 58.91 | 70.84 | 23.22 | 91.4 |
| 900 | 90 | 6.53 | 1.21 | **-1.04** | 0.64 | 453.55 | 634.3 | 910.95 | 680.72 |
| TOTAL 900 | | 2.33 | 0.45 | **-0.27** | 0.28 | 175.66 | 238.75 | 311.70 | 257.56 |
| OVERALL | | 11.2 | 1.02 | 0.66 | **0.02** | 271.53 | 332.53 | 474.36 | 438.44 |

best solution by the GA2, SS2, hyper-heuristic without naive Bayes and hyper-heuristic with naive Bayes respectively. In the Table 6, the row which starts with TOTAL 50 gives the average deviation and average CPU time for the instance with 50 nodes for different values of $p$. Similarly the row which starts with TOTAL 100 gives the average deviation and average CPU time for the instance with 100 nodes for different values of $p$. The last row in the Table 6, which starts with OVERALL gives the average deviation and average CPU time over all the instances calculated as the average of the values from rows starting with TOTAL 50, TOTAL 100. We have followed the same method of reporting in the other tables, viz. Tables 7, and 8 as well.

As shown in Table 6, on the DLPL instances our hyper-heuristic with naive Bayes achieved an average deviation of -1.21% as against the average deviation of 0.32% of GA2 and 0.02% of SS2 from the CPLEX solution. Even our hyper-heuristic approach without naive Bayes achieved an average deviation of -1.17% which is better than that of the GA2 and SS2. For instances with 50 nodes our hyper-heuristic without naive Bayes performed better than the hyper-heuristic with naive Bayes. Overall, on combining the results of instances with 50 nodes and 100 nodes of DLPL instances, our hyper-heuristic with naive Bayes outperforms our hyper-heuristic without naive Bayes.

On the OR library instances where CPLEX has reached feasible solution in Table 7, our hyper-heuristic with naive Bayes achieved an average deviation of -7.3% as against the average deviation of -5.41% of GA2 and -7.06% of SS2 from the CPLEX solution. On the same set of instances, the deviation achieved by our hyper-heuristic without naive Bayes (-7.05%) is slightly worse as compared to that of the SS2(-7.06%), but still better than that of the GA2 (-5.41%). Overall, our hyper-heuristic with naive Bayes performed better than GA2, SS2 and hyper-heuristic without naive Bayes on the instances where CPLEX has reached a feasible solution. On the OR library instances where CPLEX has not reached a feasible solution in Table 8, our hyper-heuristic approach with naive Bayes and without naive Bayes have performed better than GA2, SS2.

The effectiveness of using naive Bayes can be observed from the improved overall solution quality and reduced computation time as compared to when there is no naive Bayes classifier as can be observed from the overall average deviations and CPU times reported in Tables 6, 7 and 8. The system configuration used for executing our approach is different from the one used in [4], and hence we can't precisely compare the execution times. However, it is evident that our proposed hyper-heuristic with naive Bayes and without naive Bayes are slower in comparison to the existing approaches of GA2 and SS2.

We have also provided the instance-by-instance results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) in Appendix in seven tables. These results can be found in Tables 9, 10, 11, 12, 13, 14, and 15 of Appendix.

## 8 Conclusions

In this paper, we have proposed a hyper-heuristic based approach with naive Bayes classifier for the reliability $p$-median problem (RpMP). We have compared the results of our approach with the state-of-art approaches available in the literature [4]. The effectiveness of our approach can be observed in terms of the solution quality. Ours is the maiden hyper-heuristic approach for solving the RpMP. While ours is a greedy selection based hyper-heuristic, it serves as a motivation for other researchers to try other kind of hyper-heuristic approaches for RpMP and other facility location problems.

No approach exists in the literature that combines a machine learning technique with a hyper-heuristic approach for solving the facility location problems. Ours is the maiden approach combining the hyper-heuristic method and a machine learning technique for a facility location problem namely reliability $p$-median problem. This paves the way for other researchers to develop analogous methods using other machine learning techniques including but not limited to naive Bayes classifier, support vector machine (SVM), reinforcement learning etc. for solving the different variety of facility location problems. Further, the hyper-heuristic framework that we have developed here can be easily adopted for solving other combinatorial optimization problems also by changing the set of low level heuristics as per the problem under consideration.

## Appendix

Tables 9, 10, 11, 12, 13, 14 provide the instance by instance results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP library instances, whereas 15 provide the same on OR library instances. In the tables corresponding to DLP library instances, first column, $Dataset$, gives the name of the dataset. The second column titled CPLEX gives the objective value obtained by CPLEX. Columns 3, 4, 5 and 6 titled as GA1, GA2, GA3 and GA4 give the objective values obtained by the respective variants of the genetic algorithm as reported by [4]. Similarly, columns 7, 8, 9 and 10 titled as SS1, SS2, SS3 and SS4 give the objective values obtained by the respective variants of the scatter search as reported by [4]. Columns 11 and 12 give the objective values of our proposed hyper-heuristic approach without the naive Bayes classifier and with the naive Bayes classifier respectively. In the Table 15 corresponding to the results on OR library instances, the first column titled as $Dataset$ gives the name of the dataset. The second and third columns titled as $n$, $p$ represent the total number of nodes in the network and the number of facilities to be located respectively. The remaining columns in the Table 15 with the titles CPLEX, GA1, GA2, GA3, GA4, SS1, SS2, SS3, SS4, HH(without NB) and HH(with NB) have the same meaning as in the tables corresponding to DLP library instances. In all the Tables 9, 10, 11, 12, 13, 14 and 15, on a given instance the least objective value among all the approaches is shown in bold for easy identification.

**Table 9** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP instances with $N = 50$ and $p = 5$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | **1237.10** | 1283.99 | 1238.56 | 1717.40 | 1285.11 | 1338.13 | 1284.07 | 1666.62 | 1239.17 | **1237.10** | **1237.10** |
| gapA432 | **1285.25** | **1285.25** | **1285.25** | 1993.97 | 1453.42 | 1488.24 | 1368.70 | 2038.82 | 1406.56 | **1285.25** | **1285.25** |
| gapA532 | **1267.46** | 1352.19 | 1299.96 | 1633.49 | 1434.86 | 1374.79 | **1267.46** | 1808.13 | 1417.13 | **1267.46** | **1267.46** |
| gapA632 | **1250.02** | **1250.02** | **1250.02** | 1838.39 | 1383.45 | 1382.70 | 1287.85 | 1930.34 | 1382.70 | **1250.02** | **1250.02** |
| gapA732 | **917.77** | **917.77** | **917.77** | 1313.65 | **917.77** | **917.77** | **917.77** | 1418.62 | 920.10 | **917.77** | **917.77** |
| gapA832 | **1540.00** | 1540.60 | **1540.00** | 2029.23 | 1626.09 | 1540.31 | 1540.25 | 2032.85 | 1585.95 | **1540.00** | **1540.00** |
| gapA932 | **928.70** | **928.70** | **928.70** | 1235.65 | 970.06 | **928.70** | **928.70** | 1324.38 | 936.55 | **928.70** | **928.70** |
| gapA1032 | **1189.27** | **1189.27** | **1189.27** | 1649.87 | 1283.11 | **1189.27** | **1189.27** | 1546.23 | **1189.27** | **1189.27** | **1189.27** |
| gapA1132 | **1244.25** | **1244.25** | **1244.25** | 1710.69 | 1288.74 | **1244.25** | **1244.25** | 2002.07 | **1244.25** | **1244.25** | **1244.25** |
| gapA1232 | **1227.33** | 1317.99 | 1272.68 | 1683.47 | 1398.74 | 1298.10 | 1308.07 | 1528.97 | 1308.07 | **1227.33** | **1227.33** |
| gapA1332 | **918.28** | 930.35 | **918.28** | 1055.04 | 924.10 | 959.52 | **918.28** | 1483.31 | 1050.86 | **918.28** | **918.28** |
| gapA1432 | **916.91** | **916.91** | **916.91** | 1339.07 | 977.71 | **916.91** | **916.91** | 1483.29 | **916.91** | **916.91** | **916.91** |
| gapA1532 | **1212.90** | **1212.90** | **1212.90** | 1504.71 | 1360.55 | 1298.45 | **1212.90** | 1527.95 | 1315.90 | **1212.90** | **1212.90** |
| gapA1632 | **903.80** | **903.80** | **903.80** | 1204.20 | **903.80** | **903.80** | **903.80** | 1546.37 | **903.80** | **903.80** | **903.80** |
| gapA1732 | 1494.32 | 1494.32 | 1494.32 | 1836.40 | 1635.10 | 1494.32 | 1494.32 | 1833.50 | 1542.67 | **1494.31** | **1494.31** |
| gapA1832 | **1463.76** | **1463.76** | **1463.76** | 1787.31 | 1466.54 | **1463.76** | **1463.76** | 1876.02 | **1463.76** | **1463.76** | **1463.76** |
| gapA1932 | 889.77 | 889.77 | 889.77 | 1390.32 | 894.86 | 889.77 | 889.77 | 1385.73 | 894.86 | **889.76** | **889.76** |
| gapA2032 | 911.67 | 911.67 | 911.67 | 1251.00 | 956.11 | 911.67 | 911.67 | 1285.25 | 911.67 | **911.66** | **911.66** |

**Table 9** continued

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA2132 | 899.11 | 899.11 | 899.11 | 1205.25 | 941.23 | 899.11 | 899.11 | 1374.38 | 899.11 | **899.10** | **899.10** |
| gapA2232 | 1197.59 | 1197.59 | 1197.59 | 1597.50 | 1254.16 | 1197.59 | 1197.59 | 1746.59 | 1197.59 | **1197.58** | **1197.58** |
| gapA2332 | **1213.50** | 1229.56 | **1213.50** | 1434.67 | 1413.74 | 1260.49 | 1215.05 | 1786.83 | 1222.31 | **1213.50** | **1213.50** |
| gapA2432 | 1477.67 | 1477.67 | 1477.67 | 1552.89 | 1478.63 | 1477.67 | 1477.67 | 1554.51 | 1477.67 | **1477.66** | **1477.66** |
| gapA2532 | 1503.48 | 1598.75 | 1503.48 | 1743.13 | 1614.63 | 1643.85 | 1509.74 | 1904.36 | 1599.77 | **1503.47** | **1503.47** |
| gapA2632 | **1212.17** | **1212.17** | **1212.17** | 1627.75 | 1220.62 | 1214.62 | 1217.06 | 1591.12 | 1273.85 | **1212.17** | **1212.17** |
| gapA2732 | 1256.89 | 1388.01 | 1440.88 | 1623.91 | 1430.03 | 1527.93 | 1352.02 | 1813.22 | 1435.69 | **1256.88** | **1256.88** |
| gapA2832 | **1172.03** | **1172.03** | **1172.03** | 1501.43 | 1173.18 | **1172.03** | **1172.03** | 1588.67 | **1172.03** | **1172.03** | **1172.03** |
| gapA2932 | **1558.79** | **1558.79** | **1558.79** | 1944.96 | 1755.78 | 1682.03 | **1558.79** | 2217.70 | 1664.89 | **1558.79** | **1558.79** |
| gapA3032 | **876.71** | **876.71** | **876.71** | 1247.34 | **876.71** | **876.71** | **876.71** | 1485.36 | **876.71** | **876.71** | **876.71** |
| gapA3132 | **1222.68** | **1222.68** | **1222.68** | 1576.16 | 1267.21 | 1267.21 | **1222.68** | 1628.38 | **1222.68** | **1222.68** | **1222.68** |
| gapA3232 | **1213.80** | **1213.80** | **1213.80** | 1451.60 | 1321.91 | 1259.17 | **1213.80** | 1534.29 | 1256.39 | **1213.80** | **1213.80** |

**Table 10** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP instances with $N = 50$ and $p = 10$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | **75.45** | 79.36 | 81.70 | 89.80 | 86.30 | 83.13 | 77.57 | 98.11 | 80.96 | **75.45** | 76.92 |
| gapA432 | **66.66** | 72.80 | 67.02 | 85.58 | 83.09 | 69.46 | 66.80 | 91.62 | 76.38 | **66.66** | **66.66** |
| gapA532 | **79.41** | 81.20 | 81.42 | 93.75 | 91.79 | 80.16 | 79.55 | 94.63 | 80.27 | **79.41** | **79.41** |
| gapA632 | **72.88** | 73.05 | **72.88** | 101.88 | 89.39 | 76.48 | 73.05 | 95.49 | 79.18 | **72.88** | **72.88** |
| gapA732 | **75.55** | **75.55** | **75.55** | 88.83 | 83.54 | 75.64 | **75.55** | 93.72 | 76.76 | **75.55** | **75.55** |
| gapA832 | **91.24** | 92.26 | **91.24** | 105.93 | 110.75 | 92.36 | **91.24** | 110.45 | 91.82 | **91.24** | **91.24** |
| gapA932 | **59.43** | 60.39 | 60.57 | 79.18 | 71.33 | 61.84 | **59.43** | 85.95 | 61.99 | **59.43** | **59.43** |
| gapA1032 | **65.99** | 66.42 | **65.99** | 77.49 | 77.32 | 67.06 | **65.99** | 89.83 | 67.83 | **65.99** | **65.99** |
| gapA1132 | **85.13** | 88.85 | 86.16 | 98.67 | 98.06 | 87.91 | 85.83 | 112.27 | 85.86 | 85.23 | **85.13** |
| gapA1232 | **60.51** | 61.97 | 62.73 | 70.20 | 74.57 | 62.95 | 61.62 | 81.10 | 63.48 | **60.51** | **60.51** |
| gapA1332 | **58.87** | 60.82 | 62.13 | 69.82 | 68.10 | 62.70 | 59.67 | 80.60 | 64.55 | **58.87** | **58.87** |
| gapA1432 | **53.44** | 62.65 | 53.91 | 81.25 | 65.64 | 53.91 | **53.44** | 81.97 | 58.52 | **53.44** | **53.44** |
| gapA1532 | **59.80** | 59.81 | 59.81 | 68.65 | 72.63 | 59.81 | 59.81 | 79.65 | 59.81 | **59.80** | **59.80** |
| gapA1632 | 43.84 | 46.78 | 49.94 | 56.64 | 56.55 | 47.57 | **43.83** | 68.04 | 52.35 | 43.84 | 43.84 |
| gapA1732 | **81.50** | 86.35 | 83.85 | 87.70 | 94.05 | 84.99 | 82.14 | 96.22 | 86.87 | **81.50** | **81.50** |
| gapA1832 | **69.79** | 70.93 | **69.79** | 78.31 | 82.67 | 70.86 | **69.79** | 88.92 | 71.39 | **69.79** | **69.79** |
| gapA1932 | **58.93** | 59.56 | 60.97 | 69.65 | 66.67 | 61.45 | **58.93** | 79.27 | 59.76 | **58.93** | **58.93** |
| gapA2032 | **54.27** | 57.40 | 56.00 | 63.60 | 64.54 | 57.34 | **54.27** | 78.67 | 55.98 | **54.27** | **54.27** |
| gapA2132 | **49.73** | 51.09 | 50.83 | 63.72 | 57.18 | 55.24 | 49.91 | 62.88 | 52.90 | **49.73** | **49.73** |
| gapA2232 | **85.76** | 86.91 | 86.67 | 89.23 | 92.67 | 88.07 | **85.76** | 103.97 | 87.75 | **85.76** | **85.76** |
| gapA2332 | **69.66** | 73.82 | 70.56 | 80.28 | 82.02 | 70.91 | 69.97 | 88.79 | 73.10 | **69.66** | **69.66** |
| gapA2432 | **65.47** | 65.98 | 65.98 | 82.15 | 78.57 | 67.09 | 65.62 | 90.84 | 69.39 | **65.47** | **65.47** |
| gapA2532 | **72.97** | 75.95 | 75.72 | 93.07 | 89.00 | 75.39 | **72.97** | 94.26 | 77.55 | **72.97** | 76.52 |
| gapA2632 | **63.21** | 67.85 | **63.21** | 77.16 | 74.41 | 63.28 | **63.21** | 80.43 | 63.28 | **63.21** | **63.21** |
| gapA2732 | **60.87** | 63.26 | 63.40 | 79.53 | 69.96 | 73.68 | 62.65 | 82.57 | 69.90 | **60.87** | **60.87** |
| gapA2832 | **64.81** | 67.62 | 65.68 | 73.25 | 72.18 | 67.92 | 65.23 | 95.12 | 68.09 | **64.81** | **64.81** |
| gapA2932 | **83.14** | 88.72 | 85.70 | 97.86 | 97.88 | 90.09 | 83.18 | 99.12 | 89.13 | **83.14** | **83.14** |
| gapA3032 | 49.94 | 50.93 | **49.93** | 56.01 | 52.52 | 50.16 | **49.93** | 70.03 | 50.47 | 49.94 | 49.94 |
| gapA3132 | **79.29** | 79.52 | 80.16 | 87.64 | 86.23 | 80.53 | **79.29** | 96.56 | 80.40 | **79.29** | **79.29** |
| gapA3232 | **65.34** | 66.60 | 66.40 | 77.60 | 74.77 | 66.79 | 66.18 | 82.16 | 69.43 | **65.34** | **65.34** |

**Table 11** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP instances with $N = 50$ and $p = 15$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | 16.84 | **16.83** | **16.83** | 19.09 | 17.97 | 17.93 | **16.83** | 23.61 | 17.34 | 16.84 | 16.84 |
| gapA432 | **14.73** | 14.77 | **14.73** | 16.64 | 15.16 | 15.26 | **14.73** | 20.62 | 15.77 | **14.73** | **14.73** |
| gapA532 | 19.19 | **19.18** | **19.18** | 20.72 | 19.70 | **19.18** | **19.18** | 27.21 | 19.60 | 19.19 | 19.19 |
| gapA632 | 17.20 | **17.19** | **17.19** | 19.94 | 17.77 | **17.19** | **17.19** | 24.23 | 17.84 | 17.20 | 17.20 |
| gapA732 | **22.70** | 23.01 | 22.87 | 25.05 | 23.09 | 23.09 | 22.87 | 28.99 | 23.49 | **22.70** | **22.70** |
| gapA832 | 21.77 | 21.98 | 21.83 | 23.79 | 23.11 | 21.98 | **21.76** | 28.43 | 22.74 | 21.77 | 21.77 |
| gapA932 | 15.34 | 15.37 | **15.33** | 17.65 | 16.13 | 15.44 | **15.33** | 23.91 | 16.01 | 15.34 | 15.34 |
| gapA1032 | **18.91** | 18.92 | **18.91** | 20.58 | 19.70 | 19.20 | **18.91** | 28.08 | 19.51 | **18.91** | **18.91** |
| gapA1132 | **19.99** | 20.05 | **19.99** | 22.69 | 21.54 | **19.99** | **19.99** | 27.68 | 20.82 | **19.99** | **19.99** |
| gapA1232 | **14.30** | 14.42 | **14.30** | 16.88 | 14.63 | 14.74 | 14.42 | 19.00 | 15.60 | **14.30** | **14.30** |
| gapA1332 | **9.79** | **9.79** | **9.79** | 10.56 | 10.62 | **9.79** | **9.79** | 18.23 | 10.36 | **9.79** | **9.79** |
| gapA1432 | **18.20** | **18.20** | **18.20** | 18.78 | 18.66 | 18.39 | **18.20** | 23.86 | 18.42 | **18.20** | **18.20** |
| gapA1532 | **16.89** | **16.89** | **16.89** | 18.83 | 17.05 | **16.89** | **16.89** | 26.61 | 17.52 | **16.89** | **16.89** |
| gapA1632 | **8.78** | **8.78** | **8.78** | 10.08 | 9.22 | 9.04 | 8.90 | 13.84 | 9.04 | **8.78** | **8.78** |
| gapA1732 | **17.54** | 17.80 | 17.65 | 20.28 | 18.93 | 18.34 | 17.57 | 23.76 | 18.61 | **17.54** | **17.54** |
| gapA1832 | **13.81** | 13.94 | 13.86 | 14.64 | 14.20 | 13.90 | 13.88 | 21.21 | 14.27 | **13.81** | **13.81** |
| gapA1932 | **17.10** | **17.10** | 17.11 | 18.68 | 17.19 | **17.10** | **17.10** | 24.26 | 17.23 | **17.10** | **17.10** |
| gapA2032 | **13.01** | 13.12 | 13.04 | 13.84 | 13.48 | 13.38 | 13.09 | 16.28 | 13.35 | **13.01** | **13.01** |
| gapA2132 | **10.11** | 10.42 | 10.25 | 12.20 | 10.98 | 10.43 | **10.11** | 14.70 | 10.71 | **10.11** | **10.11** |
| gapA2232 | **20.80** | 21.29 | 21.07 | 22.70 | 21.77 | 21.63 | 21.41 | 29.62 | 22.49 | 21.07 | 21.07 |
| gapA2332 | **13.27** | 13.51 | 13.45 | 14.99 | 13.64 | 13.75 | 13.32 | 15.68 | 13.85 | **13.27** | **13.27** |
| gapA2432 | **15.17** | 15.43 | 15.41 | 17.42 | 15.66 | 16.33 | 15.19 | 21.32 | 16.66 | 15.19 | **15.17** |
| gapA2532 | **15.49** | 15.59 | **15.49** | 16.09 | 15.94 | 15.68 | 15.54 | 19.02 | 15.80 | 15.57 | 15.57 |
| gapA2632 | **15.37** | 15.41 | 15.39 | 16.11 | 15.94 | 15.43 | 15.39 | 21.82 | 15.47 | **15.37** | **15.37** |
| gapA2732 | 10.30 | 10.54 | 10.49 | 11.56 | 11.00 | 10.98 | **10.29** | 14.73 | 10.84 | 10.30 | 10.30 |
| gapA2832 | **14.48** | 14.52 | **14.48** | 15.44 | 15.23 | 15.53 | **14.48** | 21.44 | 15.09 | **14.48** | **14.48** |
| gapA2932 | **16.25** | 16.32 | **16.25** | 20.53 | 17.36 | 17.14 | 16.44 | 22.31 | 17.57 | **16.25** | **16.25** |
| gapA3032 | 9.76 | 9.76 | **9.75** | 10.48 | 9.92 | 10.14 | 9.77 | 12.60 | 10.13 | 9.76 | 9.76 |
| gapA3132 | **17.01** | 17.15 | **17.01** | 19.67 | 18.35 | 17.49 | **17.01** | 23.51 | 17.55 | **17.01** | **17.01** |
| gapA3232 | **15.22** | 15.63 | **15.22** | 18.04 | 16.24 | 15.55 | 15.33 | 25.09 | 15.65 | **15.22** | **15.22** |

**Table 12** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP instances with $N = 100$ and $p = 5$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | 4516.81 | 4647.24 | 4570.20 | 5587.38 | 4712.14 | 4694.49 | 4522.74 | 6236.50 | 4706.80 | **4516.80** | **4516.80** |
| gapA432 | **4490.92** | 4553.91 | **4490.92** | 4941.04 | 4594.73 | 4540.31 | **4490.92** | 5267.15 | **4490.92** | **4490.92** | **4490.92** |
| gapA532 | **3914.45** | **3914.45** | **3914.45** | 5013.63 | 4059.20 | 4057.91 | **3914.45** | 5366.79 | 3918.81 | **3914.45** | **3914.45** |
| gapA632 | **4797.75** | 4800.24 | 4798.65 | 5805.58 | 4891.88 | 4801.00 | 4798.65 | 5877.58 | 4813.16 | **4797.75** | **4797.75** |
| gapA732 | **4227.57** | **4227.57** | **4227.57** | 5384.74 | 4548.70 | **4227.57** | **4227.57** | 5665.65 | 4323.06 | **4227.57** | **4227.57** |
| gapA832 | 3886.61 | 3886.61 | 3886.61 | 5218.74 | 4535.45 | 4151.67 | 3886.61 | 5728.95 | 3975.77 | **3886.60** | **3886.60** |
| gapA932 | 4255.42 | 4417.76 | 4341.06 | 5403.92 | 4593.76 | 4470.37 | 4411.40 | 5480.90 | 4500.94 | **4255.41** | **4255.41** |
| gapA1032 | 4481.25 | 4555.36 | 4518.30 | 5647.46 | 4754.26 | 4747.02 | 4481.25 | 5543.76 | 4600.84 | **4481.24** | **4481.24** |
| gapA1132 | 6895.08 | 6895.08 | 6895.08 | 7575.01 | 7184.55 | 6993.69 | 6895.08 | 8134.69 | 6895.08 | **4795.00** | **4795.00** |
| gapA1232 | **3934.78** | **3934.78** | **3934.78** | 5514.56 | 4268.33 | 4036.56 | **3934.78** | 5527.87 | 4129.69 | **3934.78** | **3934.78** |

**Table 12** continued

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA1332 | 4229.22 | 4376.56 | 4229.22 | 5298.51 | 4610.39 | 4417.43 | 4321.64 | 5424.04 | 4516.19 | **4229.21** | **4229.21** |
| gapA1432 | 3907.43 | 3907.43 | 3907.43 | 4730.00 | 4136.53 | 3907.43 | 3907.43 | 5566.00 | 3968.34 | **3907.42** | **3907.42** |
| gapA1532 | 4821.97 | 4822.22 | 4857.73 | 5710.97 | 5040.41 | 4950.69 | 4830.36 | 5807.61 | 4915.57 | **4821.96** | **4821.96** |
| gapA1632 | **4200.98** | 4252.81 | **4200.98** | 5071.81 | 4493.29 | 4296.09 | 4248.68 | 5170.37 | 4295.01 | **4200.98** | **4200.98** |
| gapA1732 | 4526.37 | 4578.78 | 4529.32 | 5411.91 | 4876.02 | 4716.64 | 4529.32 | 5666.69 | 4791.24 | **4526.36** | **4526.36** |
| gapA1832 | **4485.55** | 4495.29 | **4485.55** | 5717.63 | 4676.58 | 4634.02 | 4486.61 | 5613.78 | 4501.23 | **4485.55** | **4485.55** |
| gapA1932 | 3947.21 | 3947.21 | 3947.21 | 5894.19 | 4172.39 | 4088.01 | 3947.21 | 5802.14 | 4098.75 | **3947.20** | **3947.20** |
| gapA2032 | **4522.30** | **4522.30** | **4522.30** | 5274.29 | 4734.67 | 4682.98 | **4522.30** | 5346.61 | 4703.25 | **4522.30** | **4522.30** |
| gapA2132 | **5064.98** | 5073.81 | 5065.25 | 5481.03 | 5130.33 | 5067.70 | 5065.49 | 5904.37 | 5086.18 | **5064.98** | **5064.98** |
| gapA2232 | **4266.41** | 4271.38 | **4266.41** | 6244.76 | 4593.49 | 4408.12 | **4266.41** | 5954.98 | 4543.08 | **4266.41** | **4266.41** |
| gapA2332 | 4188.36 | 4235.82 | 4188.36 | 5194.97 | 4288.59 | 4331.64 | 4188.36 | 5376.41 | 4281.22 | **4188.35** | **4188.35** |
| gapA2432 | **4191.20** | **4191.20** | **4191.20** | 4991.31 | 4347.64 | **4191.20** | **4191.20** | 5499.89 | 4248.99 | **4191.20** | **4191.20** |
| gapA2532 | **4829.47** | 5004.03 | 4880.52 | 5675.88 | 5053.57 | 4919.12 | 4959.37 | 6097.17 | 5043.23 | **4829.47** | **4829.47** |
| gapA2632 | 4231.77 | 4280.62 | 4231.77 | 5528.87 | 4739.23 | 4231.77 | 4315.68 | 5444.50 | 4501.58 | **4231.76** | **4231.76** |
| gapA2732 | **4159.29** | **4159.29** | **4159.29** | 4998.79 | 4257.98 | **4159.29** | **4159.29** | 5867.05 | 4215.74 | **4159.29** | **4159.29** |
| gapA2832 | **3881.86** | **3881.86** | **3881.86** | 4963.34 | 4237.14 | 3930.67 | **3881.86** | 5446.95 | 4026.73 | **3881.86** | **3881.86** |
| gapA2932 | **4475.80** | 4492.99 | 4477.27 | 5261.92 | 4537.41 | 4494.46 | 4478.74 | 5483.53 | 4540.14 | **4475.80** | **4475.80** |
| gapA3032 | **4216.74** | 4321.88 | 4277.02 | 5415.81 | 4412.35 | 4433.99 | 4251.67 | 5336.18 | 4361.65 | **4216.74** | **4216.74** |
| gapA3132 | **3991.38** | 4379.37 | 4132.31 | 5413.55 | 4652.70 | 4400.30 | **3991.38** | 5711.28 | 4753.36 | **3991.38** | **3991.38** |
| gapA3232 | **4514.86** | 4779.60 | **4514.86** | 5524.90 | 4879.32 | 4592.58 | 4559.72 | 5712.01 | 4679.85 | **4514.86** | **4514.86** |

**Table 13** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on DLP instances with $N = 100$ and $p = 10$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | **243.56** | 259.74 | 250.38 | 321.98 | 271.12 | 260.17 | 251.66 | 316.54 | 264.03 | **243.56** | 245.66 |
| gapA432 | **217.84** | 226.44 | 221.03 | 295.68 | 230.76 | 225.65 | 218.14 | 311.53 | 229.06 | 218.29 | **217.84** |
| gapA532 | 214.03 | 221.73 | 213.49 | 271.22 | 240.13 | 215.08 | 213.17 | 283.03 | 225.28 | **213.05** | **213.05** |
| gapA632 | 239.50 | 254.84 | 249.92 | 301.56 | 267.96 | 260.30 | 245.82 | 351.90 | 265.19 | 237.47 | **236.79** |
| gapA732 | 233.74 | 233.53 | 231.40 | 287.72 | 252.88 | 230.52 | 229.21 | 298.80 | 248.12 | **228.55** | **228.55** |
| gapA832 | **213.15** | 220.94 | 214.93 | 273.29 | 237.29 | 225.97 | 215.24 | 315.31 | 227.36 | **213.15** | **213.15** |
| gapA932 | 229.89 | 216.52 | 205.27 | 267.82 | 235.64 | 225.08 | 203.24 | 283.82 | 231.62 | **198.90** | **198.90** |
| gapA1032 | 224.84 | 225.69 | 220.59 | 282.30 | 256.23 | 227.63 | 216.99 | 289.56 | 233.56 | 217.98 | **216.83** |
| gapA1132 | 432.07 | 488.85 | 438.61 | 979.20 | 640.70 | 438.25 | 435.25 | 1234.05 | 490.88 | **229.59** | **229.59** |
| gapA1232 | 226.32 | 230.60 | 221.42 | 276.74 | 245.51 | 232.65 | 224.55 | 288.65 | 231.57 | **220.36** | **220.36** |
| gapA1332 | **228.85** | 250.59 | 236.08 | 309.74 | 269.02 | 257.53 | 230.12 | 330.00 | 259.79 | **228.85** | 232.66 |
| gapA1432 | 236.28 | 236.55 | 228.81 | 271.49 | 254.70 | 230.81 | 225.00 | 297.76 | 235.53 | **223.22** | **223.22** |
| gapA1532 | 246.56 | 258.84 | 246.66 | 298.43 | 275.00 | 257.02 | 249.43 | 315.92 | 271.61 | 246.74 | **245.67** |
| gapA1632 | 222.21 | 225.98 | 220.32 | 277.94 | 249.64 | 225.94 | 220.22 | 296.24 | 230.68 | **219.90** | **219.90** |
| gapA1732 | 236.71 | 234.97 | 230.47 | 297.11 | 264.44 | 235.33 | 230.14 | 356.72 | 234.36 | **229.74** | **229.74** |
| gapA1832 | 226.92 | 232.57 | 228.78 | 309.49 | 246.10 | 226.29 | 224.06 | 298.92 | 236.14 | **222.07** | **222.07** |
| gapA1932 | 212.39 | 234.22 | 220.96 | 282.43 | 267.26 | 235.18 | **210.16** | 323.29 | 230.62 | **210.16** | **210.16** |
| gapA2032 | **205.51** | 206.20 | 205.97 | 273.72 | 258.21 | 235.51 | **205.51** | 319.10 | 214.16 | **205.51** | **205.51** |
| gapA2132 | 244.88 | 246.94 | 238.81 | 292.61 | 264.89 | 252.67 | 236.42 | 331.49 | 248.97 | 230.94 | **230.09** |
| gapA2232 | 245.24 | 264.53 | 242.85 | 313.03 | 276.48 | 269.97 | 244.42 | 323.24 | 260.22 | **241.74** | **241.74** |

**Table 13** continued

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA2332 | **195.62** | 201.07 | 196.19 | 278.49 | 220.52 | 197.18 | 196.64 | 288.04 | 204.94 | 196.48 | 196.48 |
| gapA2432 | 200.38 | 215.61 | 197.09 | 275.17 | 232.28 | 212.23 | **195.47** | 285.20 | 215.74 | **195.47** | **195.47** |
| gapA2532 | 245.22 | 250.24 | 237.34 | 298.35 | 259.98 | 253.41 | 237.36 | 332.87 | 255.73 | 239.05 | **233.51** |
| gapA2632 | 231.35 | 242.30 | 231.29 | 279.17 | 257.04 | 239.99 | 230.67 | 295.92 | 245.78 | **229.91** | **229.91** |
| gapA2732 | 195.53 | 194.69 | 194.26 | 253.94 | 212.12 | 205.39 | **192.99** | 285.81 | 207.70 | **192.99** | **192.99** |
| gapA2832 | 201.03 | 220.35 | 207.15 | 270.40 | 229.57 | 215.26 | 207.21 | 282.45 | 224.86 | **200.70** | 201.03 |
| gapA2932 | 222.59 | 230.13 | 223.31 | 300.79 | 242.38 | 226.55 | 223.11 | 289.16 | 231.38 | **209.46** | **209.46** |
| gapA3032 | 192.87 | 200.30 | 194.12 | 276.74 | 210.66 | 194.99 | 192.57 | 273.97 | 210.65 | **192.11** | **192.11** |
| gapA3132 | 235.20 | 236.20 | 233.77 | 294.11 | 263.56 | 254.62 | 229.53 | 321.43 | 237.56 | **229.52** | **229.52** |
| gapA3232 | 242.19 | 240.92 | 228.66 | 269.96 | 255.02 | 235.76 | 228.61 | 356.79 | 247.85 | **223.99** | 225.76 |

**Table 14** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on on DLP Library instances with $N = 100$ and $p = 15$

| Dataset | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| gapA332 | 57.50 | 56.79 | 54.57 | 70.36 | 62.01 | 58.00 | **54.34** | 79.91 | 66.73 | 54.83 | 54.74 |
| gapA432 | 44.11 | 45.68 | **43.74** | 62.33 | 52.96 | 48.08 | 44.13 | 70.71 | 48.84 | 44.57 | 45.00 |
| gapA532 | **50.23** | 52.58 | 50.77 | 64.16 | 56.86 | 54.15 | 50.87 | 72.01 | 54.01 | 52.00 | 51.42 |
| gapA632 | 52.26 | 58.75 | 52.59 | 69.07 | 64.70 | 58.61 | 53.53 | 81.30 | 60.41 | 52.49 | **52.07** |
| gapA732 | 54.44 | 55.55 | 53.62 | 71.57 | 64.47 | 57.56 | 53.99 | 87.53 | 57.02 | **53.49** | **53.49** |
| gapA832 | 50.15 | 50.16 | 49.22 | 61.88 | 53.93 | 51.91 | **49.18** | 77.54 | 55.09 | 49.45 | 49.19 |
| gapA932 | 42.38 | 42.96 | 41.86 | 54.29 | 47.93 | 43.13 | 41.89 | 67.65 | 46.64 | **41.66** | 41.99 |
| gapA1032 | 53.49 | 52.27 | 50.81 | 59.64 | 58.53 | 53.85 | 51.13 | 74.88 | 57.62 | **50.50** | 50.79 |
| gapA1132 | 120.55 | 123.72 | 122.47 | 165.20 | 128.33 | 121.87 | 122.21 | 179.63 | 125.72 | **45.63** | **45.63** |
| gapA1232 | 47.63 | 48.33 | **46.50** | 61.39 | 52.01 | 52.71 | 47.01 | 71.60 | 52.53 | 47.84 | 47.71 |
| gapA1332 | 54.06 | 57.01 | 53.64 | 70.41 | 62.42 | 57.73 | **53.29** | 72.94 | 60.67 | 55.98 | 55.65 |
| gapA1432 | 62.70 | 64.59 | 62.48 | 73.65 | 70.18 | 66.70 | **62.03** | 82.95 | 67.55 | 64.23 | 63.98 |
| gapA1532 | **54.01** | 58.62 | 55.87 | 74.00 | 61.83 | 61.61 | 55.25 | 86.82 | 63.23 | 56.82 | 56.50 |
| gapA1632 | **39.44** | 45.45 | **39.44** | 57.01 | 51.47 | 46.01 | 40.11 | 64.89 | 49.36 | 42.92 | 41.33 |
| gapA1732 | 52.36 | 53.00 | **51.96** | 63.36 | 55.55 | 53.01 | 52.22 | 68.29 | 54.86 | 52.71 | 52.08 |
| gapA1832 | 43.59 | 43.48 | 42.72 | 55.40 | 50.44 | 45.32 | **42.64** | 67.30 | 47.99 | 43.10 | 42.86 |
| gapA1932 | 52.79 | 53.03 | 51.25 | 67.49 | 61.44 | 56.92 | 52.04 | 84.00 | 56.27 | 52.13 | **51.04** |
| gapA2032 | 47.85 | 49.02 | **46.68** | 58.88 | 55.86 | 48.03 | **46.68** | 70.57 | 50.72 | **46.68** | 47.50 |
| gapA2132 | 56.55 | 54.93 | 52.10 | 72.17 | 59.52 | 56.32 | 52.93 | 73.46 | 57.53 | 52.81 | **51.95** |
| gapA2232 | 56.26 | 63.42 | 57.42 | 73.94 | 65.68 | 61.23 | 56.31 | 81.42 | 63.81 | 56.25 | **56.03** |
| gapA2332 | 43.71 | 46.24 | 42.81 | 55.15 | 48.46 | 46.83 | 42.91 | 67.80 | 47.62 | 43.13 | **42.77** |
| gapA2432 | 55.20 | 55.05 | 54.05 | 65.73 | 59.63 | 56.34 | **53.61** | 89.72 | 58.55 | 54.54 | 54.38 |
| gapA2532 | 54.21 | 51.33 | **48.75** | 64.48 | 59.04 | 55.29 | 49.21 | 75.89 | 56.18 | 51.60 | 51.81 |
| gapA2632 | 52.18 | 52.88 | 50.41 | 64.13 | 57.39 | 55.09 | 50.80 | 72.80 | 56.72 | 51.49 | **49.65** |
| gapA2732 | 48.31 | 50.49 | 48.24 | 63.94 | 53.51 | 50.41 | **47.83** | 71.24 | 51.84 | 48.61 | 48.51 |
| gapA2832 | 44.38 | 45.97 | 44.37 | 61.45 | 50.62 | 44.53 | 44.42 | 69.96 | 48.49 | 44.24 | **44.03** |
| gapA2932 | 44.04 | 46.34 | 44.76 | 63.14 | 50.73 | 46.07 | 44.85 | 64.76 | 50.34 | **43.51** | **43.51** |
| gapA3032 | **42.52** | 43.35 | 42.97 | 57.59 | 47.56 | 44.51 | 43.25 | 64.38 | 47.55 | 43.06 | 42.80 |
| gapA3132 | **50.35** | 55.86 | 53.08 | 72.74 | 59.37 | 54.75 | 53.10 | 82.54 | 56.32 | **50.35** | **50.35** |
| gapA3232 | 48.49 | 49.36 | 48.08 | 62.96 | 54.20 | 50.38 | 49.27 | 66.98 | 52.66 | **47.85** | 48.33 |

**Table 15** Results of CPLEX, 4 GA variants, 4 SS variants, HH(without NB) and HH(with NB) on OR-Library instances

| Dataset | N | p | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| pmed1 | 100 | 5 | 20467.46 | 20467.46 | 20467.46 | 20514.39 | 20467.46 | 20467.46 | 20467.46 | 20670.57 | 20467.46 | **20467.45** | **20467.45** |
| pmed2 | 100 | 10 | **14351.87** | **14351.87** | **14351.87** | 14370.39 | **14351.87** | 14380.12 | 14356.24 | 15017.79 | **14351.87** | **14351.87** | **14351.87** |
| pmed3 | 100 | 20 | **5508.65** | 5611.95 | **5508.65** | 5773.99 | 5575.81 | 6087.72 | 5532.21 | 6818.61 | 5662.41 | 5577.18 | **5508.65** |
| pmed4 | 100 | 20 | **6093.05** | 6135.29 | 6152.12 | 6205.62 | 6188.57 | 6322.38 | 6146.81 | 7057.22 | 6261.37 | 6171.97 | 6140.23 |
| pmed5 | 100 | 33 | **2039.52** | 2047.50 | 2042.35 | 2051.60 | 2047.11 | 2070.89 | 2048.90 | 2201.10 | 2055.72 | 2039.92 | 2039.92 |
| pmed6 | 200 | 5 | 42308.73 | 42308.73 | 42308.73 | 42939.05 | 42308.73 | 42308.73 | 42308.73 | 43314.64 | 42308.73 | **42308.72** | **42308.72** |
| pmed7 | 200 | 10 | **31653.48** | **31653.48** | **31653.48** | 31845.36 | 31726.48 | 31728.08 | 31665.99 | 32955.19 | 31725.82 | **31653.48** | **31653.48** |
| pmed8 | 200 | 20 | 16691.89 | 15104.87 | 15066.52 | 15346.04 | 15346.80 | 15539.68 | 15042.27 | 17879.87 | 15193.30 | **15027.00** | **15027.00** |
| pmed9 | 200 | 40 | **4699.73** | 4788.92 | 4990.77 | 4768.21 | 4784.97 | 5038.05 | 4777.56 | 5642.44 | 4949.78 | 4772.37 | 4763.35 |
| pmed10 | 200 | 67 | **1878.27** | 1889.52 | 1895.12 | 1890.27 | 1931.07 | 1892.70 | 1883.59 | 2023.59 | 1897.15 | 1884.78 | 1895.03 |
| pmed11 | 300 | 5 | 65875.01 | 65875.01 | 65875.01 | 66753.71 | 65875.01 | 65875.01 | 65875.01 | 67876.69 | 65875.01 | **65875.00** | **65875.00** |
| pmed12 | 300 | 10 | **49268.65** | **49268.65** | **49268.65** | 49723.46 | 49581.42 | 49665.09 | **49268.65** | 52577.82 | 49288.13 | **49268.65** | **49268.65** |
| pmed13 | 300 | 30 | 90000 | 13783.50 | 14638.50 | 14445.00 | 13986.00 | 14701.50 | **13725.00** | 18432.00 | 14377.50 | 13904.64 | 13917.30 |
| pmed14 | 300 | 60 | 7392.94 | 5232.41 | 5817.91 | 5337.85 | 5644.01 | 5393.46 | 5056.99 | 5887.95 | 5131.14 | 5094.71 | **5015.71** |
| pmed15 | 300 | 100 | **2244.28** | 2485.21 | 2560.70 | 2530.68 | 2626.79 | 2364.50 | 2250.21 | 2609.93 | 2284.56 | 2286.71 | 2247.06 |
| pmed16 | 400 | 5 | 88031.1 | 88031.10 | 88031.10 | 91128.12 | 88031.10 | 88031.10 | 88031.10 | 91384.34 | 88048.61 | **88031.08** | **88031.08** |
| pmed17 | 400 | 10 | 67945.77 | 67764.19 | 67764.19 | 69131.71 | 67865.78 | 67992.41 | 67774.26 | 71713.56 | 67814.61 | **67764.18** | **67764.18** |
| pmed18 | 400 | 40 | 120000 | 13062.00 | 15762.00 | 13152.00 | 14802.00 | 14418.00 | 13086.00 | 16296.00 | 13290.00 | 13495.83 | **12877.35** |
| pmed19 | 400 | 80 | 120000 | 5286.00 | 5898.00 | 5328.00 | 5874.00 | 5148.00 | 4632.00 | 5310.00 | 4752.00 | **4600.15** | 4627.38 |
| pmed20 | 400 | 133 | 2950.44 | 2901.07 | 2500.58 | 3001.28 | 2666.98 | 2407.53 | 2455.08 | 2643.90 | 2448.55 | 2383.04 | **2363.95** |
| pmed21 | 500 | 5 | 112869.05 | 112869.05 | 112869.05 | 114347.63 | 112869.05 | 112869.05 | 112869.05 | 115449.87 | 112869.05 | 112869.03 | 112869.03 |
| pmed22 | 500 | 10 | 150000 | 86527.50 | 86550.00 | 87900.00 | 86760.00 | 86655.00 | **86437.50** | 92632.50 | 86617.50 | 86644.45 | 86644.45 |
| pmed23 | 500 | 50 | 150000 | 14190.00 | 14100.00 | 14070.00 | 14812.50 | 13207.50 | 13005.00 | 15300.00 | 13665.00 | **12799.07** | 12814.98 |
| pmed24 | 500 | 100 | 150000 | 6060.00 | 6375.00 | 6150.00 | 6787.50 | 5407.50 | 5032.50 | 5415.00 | 5077.50 | **4843.08** | 4916.49 |
| pmed25 | 500 | 167 | 150000 | 3307.50 | 2640.00 | 3037.50 | 2797.50 | 2647.50 | **2512.50** | 2827.50 | 2535.00 | 2578.90 | 2567.17 |
| pmed26 | 600 | 5 | 180000 | 137403.00 | 137403.00 | 138564.00 | 137403.00 | 137403.00 | 137403.00 | 140814.00 | 137403.00 | 137404.28 | 137404.28 |
| pmed27 | 600 | 10 | 180000 | 105507.00 | 105489.00 | 107757.00 | 105588.00 | 105642.00 | 105003.00 | 111888.00 | 105183.00 | 104996.45 | 104996.45 |
| pmed28 | 600 | 60 | 180000 | 14400.00 | 14472.00 | 14292.00 | 14958.00 | 12762.00 | 13050.00 | 14418.00 | 12915.00 | 12542.69 | **12344.55** |
| pmed29 | 600 | 120 | 180000 | 6489.00 | 6417.00 | 6390.00 | 7200.00 | 5679.00 | 5211.00 | 5733.00 | 5202.00 | 5239.54 | **5024.81** |
| pmed30 | 600 | 200 | 180000 | 3474.00 | 3294.00 | 3528.00 | 3834.00 | 3276.00 | 2664.00 | 3222.00 | 2826.00 | **2630.77** | 2633.19 |
| pmed31 | 700 | 5 | 210000 | 161049.00 | 161049.00 | 163327.50 | 161049.00 | 161049.00 | 161049.00 | 165690.00 | 161049.00 | 161058.58 | 161058.58 |

**Table 15** continued

| Dataset | N | p | CPLEX | GA1 | GA2 | GA3 | GA4 | SS1 | SS2 | SS3 | SS4 | HH (without NB) | HH (with NB) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| pmed32 | 700 | 10 | 210000 | 123301.50 | 123417.00 | 125947.50 | 123921.00 | 123637.50 | 123175.50 | 130431.00 | 123427.50 | 123501.52 | 124046.65 |
| pmed33 | 700 | 70 | 210000 | 15351.00 | 13765.50 | 15603.00 | 14395.50 | 12747.00 | 12946.50 | 14815.50 | 13020.00 | 12476.10 | **12355.45** |
| pmed34 | 700 | 140 | 210000 | 6457.50 | 6699.00 | 6814.50 | 7203.00 | 6037.50 | 5176.50 | 6100.50 | 5208.00 | **5174.56** | **5174.56** |
| pmed35 | 800 | 5 | 240000 | 186228.00 | 186228.00 | 188844.00 | 186360.00 | 186516.00 | 186228.00 | 190764.00 | 186504.00 | 186221.31 | 186221.31 |
| pmed36 | 800 | 10 | 240000 | 145812.00 | 145548.00 | 147996.00 | 146268.00 | 145980.00 | 145068.00 | 153648.00 | 145320.00 | 145061.95 | 145061.95 |
| pmed37 | 800 | 80 | 240000 | 16260.00 | 16476.00 | 15960.00 | 17388.00 | 14244.00 | 13440.00 | 15108.00 | 13560.00 | 13802.09 | **13055.55** |
| pmed38 | 900 | 5 | 270000 | 208102.50 | 208102.50 | 210492.00 | 208116.00 | 208102.50 | 208102.50 | 214056.00 | 208156.50 | 208091.33 | 208091.33 |
| pmed39 | 900 | 10 | 270000 | 161527.50 | 161689.50 | 164997.00 | 162337.50 | 161487.00 | 161176.50 | 173056.50 | 161176.50 | 161339.59 | 161319.81 |
| pmed40 | 900 | 90 | 270000 | 16497.00 | 14310.00 | 16807.50 | 14634.00 | 14269.50 | 13594.50 | 15808.50 | 13581.00 | **13292.20** | 13517.91 |

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**Data Availability** All the test instances used in this paper and their detailed results will be made available through e-mail from the corresponding author.

## Declarations

**Ethical approval** This article does not contain any studies with human participants or animals performed by any of the authors.

**Conflicts of interest** The authors declare that they have no conflict of interest.

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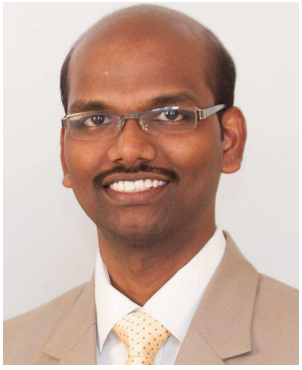

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